

Statistics

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Statistics is the science of making informed decisions. It is an area of mathematics that is widely applicable, and covers the design of experiments, the analysis of data, statistical inference, and the communication of uncertainty and risk. In this course we will focus on formal statistical inference.

Probability, as we met it in IA, starts from a fully specified model and asks what the data will look like. Statistics runs that arrow backwards: we are handed the data, and asked what the model must have been. That inversion is what gives the subject its character – there is never a single right answer, only estimates with quantified uncertainty, and procedures whose long-run behaviour we can guarantee. Almost everything in these notes is an attempt to make one of the words ‘estimate’, ‘uncertainty’ or ‘guarantee’ precise.

This article constitutes my notes for the ‘Statistics’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered. Note that familiarity with classical probability theory is assumed, and a full recap is not included – the reader is referred to my ‘Probability’ notes for anything in Section 1 that is unfamiliar.

Contents

§1 Introduction

§1.1 Parametric Inference

Let $X_1, \dots, X_n$ be i.i.d random variables. We will assume that the distribution of X_i belongs to some family with parameter $\theta \in \Theta$. For example, we could have $X_1 \sim \text{Poisson}(\mu)$ with $\theta = \mu \in \Theta = (0, \infty)$. We can also have a multi-dimensional parameter such as $X_1 \sim N(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2) \in \Theta = \mathbb{R} \times (0, \infty)$. We will use the observed $X = (X_1, \dots, X_n)$ to make inferences about the parameter θ . Such inferences include:

1. Point estimate of $\hat{\theta}(X)$ of θ ;
2. An interval estimate $(\hat{\theta}_1(X), \hat{\theta}_2(X))$ of θ ;
3. Testing hypothesis about θ , that is, checking whether there is evidence in the data X against a given hypothesis.

In this course we will usually assume that the distribution family of $X_1, \dots, X_n$ is known, and that the parameter is unknown. From time to time we will be able to make slightly more general statements.

This setup is called **parametric** inference, because the unknown is a finite-dimensional parameter. It is a genuine modelling assumption, and a strong one: if the family is badly chosen then no amount of clever estimation will save us. In practice one checks the family afterwards, and Section 5 gives us the tools (goodness-of-fit tests) to do exactly that.

§1.2 Random Samples and Statistics

We should fix some language before we start.

Definition 1.1 (Random Sample)

A **random sample** of size n from a distribution f is a collection $X = (X_1, \dots, X_n)$ of independent random variables, each with distribution f . We write $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$.

Definition 1.2 (Statistic)

A **statistic** is any function $T(X)$ of the data alone. In particular a statistic may *not* depend on the unknown parameter θ : it must be something we can actually compute once the data are in front of us.

The distribution of a statistic $T(X)$ – which of course will in general depend on θ – is called its **sampling distribution**. Most of statistics consists of choosing a statistic whose sampling distribution we understand well enough to say something useful.

The two statistics we will use over and over are the sample mean and the sum of squared deviations from it.

Definition 1.3 (Sample Mean and S_{XX})

Given a sample $X_1, \dots, X_n$ we write

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2.$$

If $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2$ then linearity of expectation and bilinearity of variance immediately give

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n},$$

so the sample mean is ‘centred at the right place’ and gets more concentrated as n grows. It is worth recording the standardised version

$$Z_n = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma},$$

which has mean 0 and variance 1 whatever μ and σ are. The central limit theorem says Z_n is approximately $N(0, 1)$ for large n , and this single fact is responsible for a remarkable fraction of the approximations in this course.

There is one small wrinkle worth noticing now, since it will bite us later. A short computation gives

$$\mathbb{E}[S_{XX}] = \mathbb{E} \left[\sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \right]$$

$$= n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2,$$

so S_{XX}/n underestimates σ^2 , and it is $S_{XX}/(n-1)$ that is unbiased. The intuition is that we have used up one ‘degree of freedom’ estimating the mean, and the data look slightly less spread out about $\bar{X}$ than about the true μ . This phrase ‘degrees of freedom’ will be given a precise geometric meaning in Section 7.

§1.3 Sampling Distributions from the Normal

Almost all the exact distribution theory in this course comes from one place: sums of squares of independent standard normals. We collect the three families here, since we will want to refer to them from the very beginning. All three are already implicitly present in IA Probability as special cases or ratios of gamma distributions.

Recall that $X \sim \text{Gamma}(\alpha, \lambda)$ has density proportional to $x^{\alpha-1}e^{-\lambda x}$ on $(0, \infty)$, and that gammas with a common rate add: $\text{Gamma}(\alpha_1, \lambda) + \text{Gamma}(\alpha_2, \lambda) = \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$ for independent summands.

Definition 1.4 (Chi-Squared Distribution)

Let $Z_1, \dots, Z_k$ be i.i.d $N(0, 1)$. Then

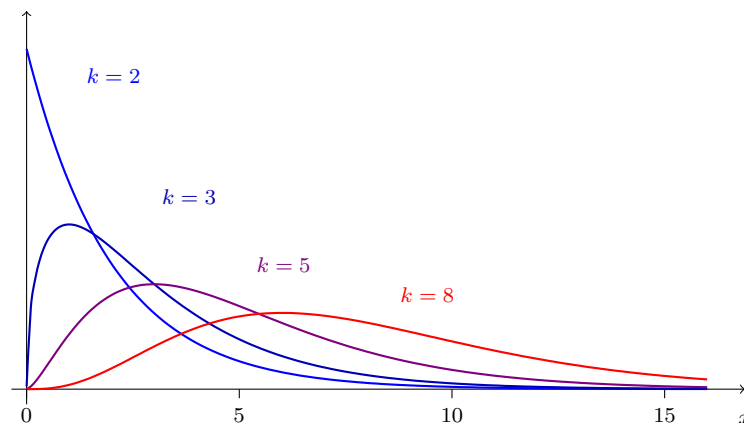
$$\chi_k^2 := Z_1^2 + \dots + Z_k^2$$

is said to have the **chi-squared distribution on k degrees of freedom**.

If $Z \sim N(0, 1)$ then $Z^2 \sim \text{Gamma}(1/2, 1/2)$ (a one-line change of variables), so by the addition rule $\chi_k^2 \sim \text{Gamma}(k/2, 1/2)$, with density

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}, \quad x > 0.$$

In particular $\mathbb{E}[\chi_k^2] = k$ and $\text{Var}(\chi_k^2) = 2k$. The shape changes character as k increases: for $k \leq 2$ the density is decreasing, and for $k \geq 3$ it has an interior mode at $k-2$, drifting rightwards and flattening out.



We will write $\chi_k^2(\alpha)$ for the **upper α point** of the distribution, that is, the number with $\mathbb{P}(\chi_k^2 > \chi_k^2(\alpha)) = \alpha$. The same convention will be used for every distribution in these notes.

Definition 1.5 (Student's t -Distribution)

Let $Z \sim N(0, 1)$ and $V \sim \chi_k^2$ be independent. Then

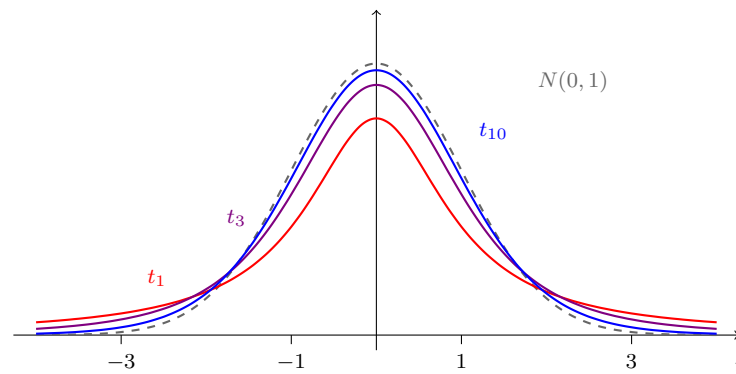
$$T = \frac{Z}{\sqrt{V/k}}$$

is said to have the **t -distribution on k degrees of freedom**, written $T \sim t_k$.

The point of this definition – which looks arbitrary until you see it used – is that it describes what happens when you standardise a normal mean but have to *estimate* the standard deviation rather than knowing it. We will prove exactly this in Section 6. One can compute the density to be

$$f_T(t) = \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2}) \sqrt{\pi k}} \left(1 + \frac{t^2}{k}\right)^{-(k+1)/2},$$

which is symmetric and bell-shaped, but with polynomially decaying (hence much fatter) tails than the normal. Indeed $\mathbb{P}(T > t) > \mathbb{P}(Z > t)$ for every $t > 0$: having to estimate σ costs us, and the cost shows up precisely as extra probability far from the centre.



As $k \rightarrow \infty$ we have $V/k \rightarrow 1$ by the law of large numbers, so $t_k \rightarrow N(0, 1)$; the picture above shows this happening already by $k = 10$. For small k the tails really are heavy: t_1 is the Cauchy distribution, which has no mean at all. In general $\mathbb{E}[T] = 0$ for $k > 1$ and $\text{Var}(T) = k/(k-2)$ for $k > 2$, with the variance infinite at $k = 2$ and undefined at $k = 1$.

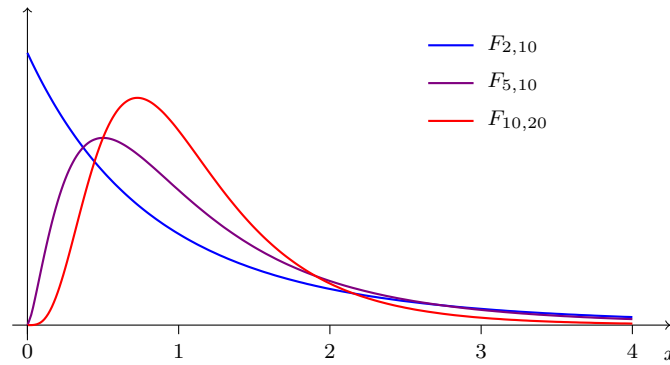
Definition 1.6 (F -Distribution)

Let $V \sim \chi_m^2$ and $W \sim \chi_n^2$ be independent. Then

$$F = \frac{V/m}{W/n}$$

is said to have the **F -distribution on m and n degrees of freedom**, written $F \sim F_{m,n}$.

This is what appears when we want to compare two variances, or – much more importantly – when we want to compare the fit of two nested models by comparing two sums of squares. It is supported on $(0, \infty)$ and concentrates around 1 as both $m, n \rightarrow \infty$, which is what one would expect of a ratio of two estimates of the same quantity.



Two relations are worth remembering, both immediate from the definitions. First, if $T \sim t_n$ then $T^2 \sim F_{1,n}$, since $T^2 = Z^2/(V/n)$ and $Z^2 \sim \chi_1^2$. Second, if $F \sim F_{m,n}$ then $1/F \sim F_{n,m}$. The first of these will let us identify a two-sided t -test with an F -test in Section 7.

§2 Estimation

Suppose $X_1, \dots, X_n$ are i.i.d observations with a pdf (or pmf) $f_X(x \mid \theta)$ where θ is an unknown parameter in a parameter space Θ . The first and most basic inferential question is: what single value of θ do the data suggest?

§2.1 Estimators, Bias and Mean Squared Error

Definition 2.1 (Estimator)

An **estimator** is a statistic or function of the data $T(X) = \hat{\theta}$ which does *not* depend on θ , and is used to approximate the true parameter θ .

The distribution of $T(X)$ is its **sampling distribution**.

We can then go on to define the *bias* of an estimator.

Definition 2.2 (Bias)

The **bias** of $\hat{\theta} = T(X)$ is

$$\text{bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta,$$

where $\mathbb{E}_\theta$ gives expectation in the model $X_i \sim f_X(\cdot \mid \theta)$.

Remark. In general the bias is a function of the true parameter θ , even though it is not explicit in the notation $\text{bias}(\hat{\theta})$.

Definition 2.3 (Unbiased Estimator)

We say that $\hat{\theta}$ is **unbiased** if $\text{bias}(\hat{\theta}) = 0$ for *all* values of the true parameter θ .

Example 2.4 (Defining an Estimator)

Consider the i.i.d random variables $X_1, \dots, X_n \sim N(\mu, 1)$. Then we can define the

estimator

$$\hat{\mu} = T(X) = \frac{1}{n} \sum_i X_i = \bar{X}.$$

The sampling distribution of $\hat{\mu}$ is $T(X) \sim N(\mu, 1/n)$.

We can see that $\hat{\mu}$ is an unbiased estimator since

$$\mathbb{E}_\theta(\hat{\mu}) = \mathbb{E}_\mu(\bar{X}) = \mu,$$

for all $\mu \in \mathbb{R}$.

Unbiasedness on its own is a very weak requirement – it says nothing about how far $\hat{\theta}$ typically is from θ , only that the errors average out. One way of measuring ‘how far’ $\hat{\theta}$ is from θ ‘on average’ is given by the *mean squared error* (mse).

Definition 2.5 (Mean Squared Error)

The **mean squared error** of $\hat{\theta}$ is defined to be

$$\text{mse}(\hat{\theta}) = \mathbb{E}_\theta \left[(\hat{\theta} - \theta)^2 \right].$$

Again, note that $\text{mse}(\hat{\theta})$ may depend on θ , even though this dependence is omitted from the notation.

Proposition 2.6 (Bias-Variance Decomposition)

For any estimator $\hat{\theta}$ with finite second moment,

$$\text{mse}(\hat{\theta}) = \text{Var}_\theta(\hat{\theta}) + \text{bias}^2(\hat{\theta}).$$

Proof. We add and subtract $\mathbb{E}_\theta \hat{\theta}$ inside the square:

$$\begin{aligned} \text{mse}(\hat{\theta}) &= \mathbb{E}_\theta \left[\left(\hat{\theta} - \mathbb{E}_\theta \hat{\theta} + \mathbb{E}_\theta \hat{\theta} - \theta \right)^2 \right] \\ &= \mathbb{E}_\theta \left[\left(\hat{\theta} - \mathbb{E}_\theta \hat{\theta} \right)^2 \right] + 2 \left(\mathbb{E}_\theta \hat{\theta} - \theta \right) \mathbb{E}_\theta \left[\hat{\theta} - \mathbb{E}_\theta \hat{\theta} \right] + \left(\mathbb{E}_\theta \hat{\theta} - \theta \right)^2 \\ &= \underbrace{\text{Var}_\theta(\hat{\theta})}_{\geq 0} + \underbrace{\text{bias}_\theta^2(\hat{\theta})}_{\geq 0}, \end{aligned}$$

since the cross term vanishes: $\mathbb{E}_\theta[\hat{\theta} - \mathbb{E}_\theta \hat{\theta}] = 0$. □

This decomposition is the single most useful identity in estimation theory, because it shows that there is a genuine *tradeoff*: we may be able to buy a large reduction in variance at the cost of a small bias, and come out ahead. Insisting on unbiasedness forbids that trade entirely. Let’s see this happen.

Example 2.7 (A Biased Estimator that Wins)

Consider the random variable $X \sim \text{Bin}(n, \theta)$, and suppose n is known, with $\theta \in [0, 1]$ being an unknown parameter.

The standard estimator for this example would be

$$T_U = \frac{X}{n},$$

which can be thought of as the ‘proportion of successes observed’. This estimator is clearly unbiased since

$$\mathbb{E}_\theta(T_U) = \frac{\mathbb{E}_\theta X}{n} = \frac{n\theta}{n} = \theta.$$

We can also compute that

$$\text{mse}(T_U) = \text{Var}_\theta(T_U) = \frac{\theta(1-\theta)}{n}.$$

Now consider another estimator

$$T_B = \frac{X+1}{n+2} = w \cdot \underbrace{\frac{X}{n}}_{T_U} + (1-w) \cdot \frac{1}{2}, \quad w = \frac{n}{n+2},$$

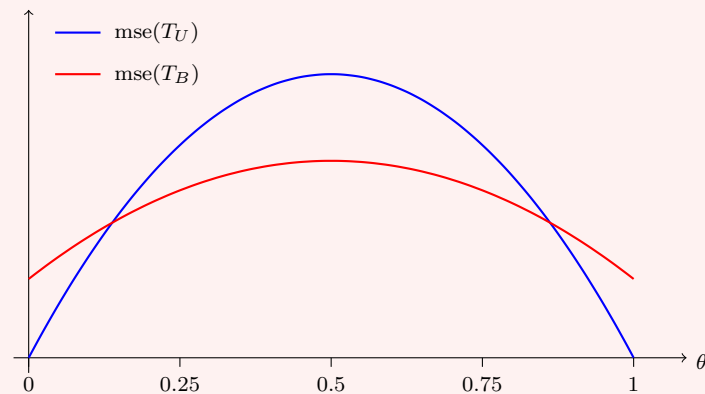
which shrinks the observed proportion towards $1/2$. Its bias is

$$\text{bias}(T_B) = \frac{n\theta + 1}{n+2} - \theta = \frac{1-2\theta}{n+2},$$

which vanishes only at $\theta = 1/2$, and its variance is $w^2\theta(1-\theta)/n$. So

$$\text{mse}(T_B) = w^2 \cdot \frac{\theta(1-\theta)}{n} + (1-w)^2 \left(\frac{1}{2} - \theta\right)^2.$$

Plotting these against θ for $n = 10$, we see that neither estimator dominates.



The biased estimator is better over a broad range of θ near $1/2$, and worse near the endpoints. Which one we prefer therefore depends on what we believe about θ before seeing the data – an observation which will lead us, in Section 4, straight to Bayesian inference.

It is also worth seeing that insisting on unbiasedness can produce estimators that are frankly absurd.

Example 2.8 (Unbiasedness Gone Wrong)

Suppose $X \sim \text{Poisson}(\lambda)$ and we wish to estimate $\theta = \mathbb{P}(X = 0)^2 = e^{-2\lambda}$. For $T(X)$ to be unbiased we need, for all $\lambda > 0$,

$$\mathbb{E}_\lambda[T(X)] = \sum_{x=0}^{\infty} T(x) \frac{\lambda^x e^{-\lambda}}{x!} = e^{-2\lambda},$$

that is, $\sum_{x \geq 0} T(x) \lambda^x / x! = e^{-\lambda}$. Comparing coefficients of λ^x with the power series for $e^{-\lambda}$ gives $T(x) = (-1)^x$.

So the *unique* unbiased estimator of θ takes only the values ± 1 , and estimates a quantity known to lie in $(0, 1)$ by -1 whenever we observe an odd number of events. Unbiasedness is clearly not, by itself, a guarantee of anything sensible.

§2.2 Sufficiency

The data $X = (X_1, \dots, X_n)$ typically contains far more detail than we need. If $X_1, \dots, X_n$ are i.i.d Bernoulli, it seems clear that the *order* of the successes tells us nothing about θ ; only the total number matters. Sufficiency is the notion that makes this precise.

Definition 2.9 (Sufficient Statistic)

A statistic $T(X)$ is **sufficient** for θ if the conditional distribution of X given $T(X)$ does not depend on θ .

In other words – once we know $T(X)$, the remaining randomness in X is pure noise, generated by a mechanism we could simulate ourselves without knowing θ . Nothing about θ is lost by throwing away X and keeping only $T(X)$. Note that θ and $T(X)$ may both be vector-valued, and need not have the same dimension.

Example 2.10 (Bernoulli Trials)

Let $X_1, \dots, X_n$ be i.i.d Bernoulli(θ) with $\theta \in [0, 1]$. The joint mass function is

$$f_X(x | \theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i},$$

which depends on x only through $t = \sum_i x_i$. Let $T(X) = \sum_i X_i$; since $T \sim \text{Bin}(n, \theta)$, for any x with $\sum_i x_i = t$ we get

$$\mathbb{P}_\theta(X = x | T(X) = t) = \frac{\mathbb{P}_\theta(X = x)}{\mathbb{P}_\theta(T = t)} = \frac{\theta^t (1 - \theta)^{n-t}}{\binom{n}{t} \theta^t (1 - \theta)^{n-t}} = \frac{1}{\binom{n}{t}},$$

and the probability is 0 if $\sum_i x_i \neq t$. This is free of θ , so T is sufficient: given the total, the particular arrangement of successes is uniform over all $\binom{n}{t}$ arrangements, regardless of θ .

Checking sufficiency from the definition means computing a conditional distribution, which is tedious. Happily there is a criterion that lets us read sufficiency straight off the likelihood.

Theorem 2.11 (Factorisation Criterion)

A statistic T is sufficient for θ if and only if we can write

$$f_X(x | \theta) = g(T(x), \theta) h(x)$$

for some functions g and h , where h does not depend on θ .

Proof. We give the proof in the discrete case; the continuous case is identical with sums replaced by integrals (and a little care about null sets).

Suppose first that the factorisation holds, and let $t = T(x)$. Then

$$\begin{aligned} \mathbb{P}_\theta(X = x | T(X) = t) &= \frac{\mathbb{P}_\theta(X = x, T(X) = t)}{\mathbb{P}_\theta(T(X) = t)} \\ &= \frac{g(T(x), \theta) h(x)}{\sum_{x': T(x') = t} g(T(x'), \theta) h(x')} \\ &= \frac{g(t, \theta) h(x)}{g(t, \theta) \sum_{x': T(x') = t} h(x')} = \frac{h(x)}{\sum_{x': T(x') = t} h(x')}, \end{aligned}$$

which does not depend on θ . (If $T(x) \neq t$ the conditional probability is 0, which certainly does not depend on θ .) So T is sufficient.

Conversely, suppose T is sufficient. Then

$$\begin{aligned} f_X(x | \theta) &= \mathbb{P}_\theta(X = x) = \mathbb{P}_\theta(X = x, T(X) = T(x)) \\ &= \underbrace{\mathbb{P}_\theta(X = x | T(X) = T(x))}_{=h(x)} \underbrace{\mathbb{P}_\theta(T(X) = T(x))}_{=g(T(x), \theta)}, \end{aligned}$$

where the first factor is free of θ precisely by sufficiency. $\square$

Let's put this to work.

Example 2.12 (Bernoulli, Again)

With $f_X(x | \theta) = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}$ we may take $T(x) = \sum_i x_i$, $g(t, \theta) = \theta^t (1 - \theta)^{n - t}$ and $h(x) \equiv 1$. So T is sufficient, and we never had to compute a conditional distribution.

Example 2.13 (Uniform on $[0, \theta]$)

Let $X_1, \dots, X_n$ be i.i.d $U[0, \theta]$ with $\theta > 0$. The joint density is

$$f_X(x | \theta) = \prod_{i=1}^n \frac{1}{\theta} \mathbb{1}\{0 \leq x_i \leq \theta\} = \underbrace{\frac{1}{\theta^n} \mathbb{1}\left\{\max_i x_i \leq \theta\right\}}_{g(T(x), \theta)} \cdot \underbrace{\mathbb{1}\left\{\min_i x_i \geq 0\right\}}_{h(x)},$$

so $T(X) = \max_i X_i$ is sufficient for θ . This is a good example of a sufficient statistic that is not a sum: only the largest observation constrains θ at all.

Example 2.14 (Normal with Both Parameters Unknown)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$ unknown. Expanding the exponent,

$$f_X(x | \theta) = (2\pi\sigma^2)^{-n/2} \exp \left(-\frac{1}{2\sigma^2} \sum_i x_i^2 + \frac{\mu}{\sigma^2} \sum_i x_i - \frac{n\mu^2}{2\sigma^2} \right),$$

which depends on x only through $(\sum_i x_i, \sum_i x_i^2)$. Taking $h \equiv 1$, this pair is sufficient for (μ, σ^2) .

§2.3 Minimal Sufficiency

Sufficient statistics are far from unique. The whole data $T(X) = X$ is always sufficient, and any injective function of a sufficient statistic is again sufficient. What we actually want is a sufficient statistic that compresses as much as possible.

Definition 2.15 (Minimal Sufficiency)

A sufficient statistic T is **minimal sufficient** if it is a function of every other sufficient statistic; equivalently, if S is sufficient then $S(x) = S(y)$ implies $T(x) = T(y)$.

Remark. If S and T are minimal sufficient then they are in bijection, that is, $T(x) = T(y)$ if and only if $S(x) = S(y)$. So we can think of minimal sufficient statistics as being unique up to bijection.

Theorem 2.16 (Criterion for Minimal Sufficiency)

Suppose that $f_X(x | \theta)/f_X(y | \theta)$ is constant in θ if and only if $T(x) = T(y)$. Then T is minimal sufficient.

Before proving this, it is worth understanding what the hypothesis says. Write $x \sim_1 y$ if the likelihood ratio $f_X(x | \theta)/f_X(y | \theta)$ is free of θ , and $x \sim_2 y$ if $T(x) = T(y)$. Both are equivalence relations on the sample space, and the hypothesis is exactly that they have the same equivalence classes. Note that this also gives a *recipe*: compute the classes of $\sim_1$, and define T to be any labelling of them. So minimal sufficient statistics always exist.

Proof. We first show T is sufficient. For each value t in the range of T pick a representative z_t with $T(z_t) = t$. Then for any x ,

$$f_X(x | \theta) = \underbrace{f_X(z_{T(x)} | \theta)}_{g(T(x), \theta)} \cdot \underbrace{\frac{f_X(x | \theta)}{f_X(z_{T(x)} | \theta)}}_{h(x)}.$$

The first factor depends on x only through $T(x)$, and the second is free of θ by hypothesis, since $T(x) = T(z_{T(x)})$. By the factorisation criterion, T is sufficient.

Now let S be any other sufficient statistic, so $f_X(x | \theta) = g_S(S(x), \theta)h_S(x)$. Suppose

$S(x) = S(y)$. Then

$$\frac{f_X(x | \theta)}{f_X(y | \theta)} = \frac{g_S(S(x), \theta) h_S(x)}{g_S(S(y), \theta) h_S(y)} = \frac{h_S(x)}{h_S(y)},$$

which is constant in θ . By hypothesis this forces $T(x) = T(y)$. So T is a function of S , and as S was arbitrary, T is minimal sufficient. $\square$

Example 2.17 (Normal Samples)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$ with both parameters unknown. Then

$$\frac{f_X(x | \mu, \sigma^2)}{f_X(y | \mu, \sigma^2)} = \exp \left(-\frac{1}{2\sigma^2} \left(\sum_i x_i^2 - \sum_i y_i^2 \right) + \frac{\mu}{\sigma^2} \left(\sum_i x_i - \sum_i y_i \right) \right).$$

This is constant in (μ, σ^2) if and only if both bracketed differences vanish, that is, if and only if $\sum_i x_i = \sum_i y_i$ and $\sum_i x_i^2 = \sum_i y_i^2$. Hence

$$T(X) = \left(\sum_i X_i, \sum_i X_i^2 \right)$$

is minimal sufficient, as is the more familiar bijective repackaging $(\bar{X}, S_{XX})$.

Example 2.18 (Dimension Mismatch)

The parameter and a minimal sufficient statistic need not have the same dimension. Take $X_1, \dots, X_n \sim N(\mu, \mu^2)$, so $\Theta = \mathbb{R} \setminus \{0\}$ is one-dimensional. Running the same computation with $\sigma^2 = \mu^2$, the ratio is

$$\exp \left(-\frac{1}{2\mu^2} \left(\sum_i x_i^2 - \sum_i y_i^2 \right) + \frac{1}{\mu} \left(\sum_i x_i - \sum_i y_i \right) \right),$$

and since $1/\mu$ and $1/\mu^2$ are not proportional as functions of μ , this is constant only when both differences vanish. So $(\bar{X}, S_{XX})$ is still minimal sufficient: a two-dimensional statistic for a one-dimensional parameter.

§2.4 The Rao-Blackwell Theorem

We now have a notion of ‘the data, compressed without loss’. The Rao-Blackwell theorem says that this compression is not merely lossless, but actively *useful*: averaging any estimator over the discarded information can only improve it.

From now on we will usually drop the subscript θ from $\mathbb{E}_\theta$ and $\mathbb{P}_\theta$, with the understanding that all expectations are taken in the model with true parameter θ .

Theorem 2.19 (Rao-Blackwell)

Let T be sufficient for θ , and let $\tilde{\theta}$ be an estimator with $\mathbb{E}[\tilde{\theta}^2] < \infty$ for all θ . Define

$$\hat{\theta} = \mathbb{E}[\tilde{\theta} \mid T].$$

Then $\hat{\theta}$ is an estimator, and for all θ ,

$$\mathbb{E}[(\hat{\theta} - \theta)^2] \leq \mathbb{E}[(\tilde{\theta} - \theta)^2],$$

with equality if and only if $\tilde{\theta}$ is (almost surely) a function of T .

Remark. It is essential that T be *sufficient*, and not merely any statistic. Otherwise $\hat{\theta}$ would generally depend on θ and so fail to be an estimator at all:

$$\hat{\theta} = \mathbb{E}[\tilde{\theta} \mid T] = \int \tilde{\theta}(x) \underbrace{f_{X|T}(x \mid T)}_{\text{free of } \theta \text{ as } T \text{ is sufficient}} dx.$$

Sufficiency is exactly the condition that makes this conditional average computable from the data.

Proof. By the tower property,

$$\mathbb{E}[\hat{\theta}] = \mathbb{E}[\mathbb{E}[\tilde{\theta} \mid T]] = \mathbb{E}[\tilde{\theta}],$$

so $\hat{\theta}$ and $\tilde{\theta}$ have exactly the same bias. By the conditional variance formula,

$$\text{Var}(\tilde{\theta}) = \mathbb{E} \left[\underbrace{\text{Var}(\tilde{\theta} \mid T)}_{\geq 0} \right] + \underbrace{\text{Var}(\mathbb{E}[\tilde{\theta} \mid T])}_{=\text{Var}(\hat{\theta})} \geq \text{Var}(\hat{\theta}).$$

Now apply the bias-variance decomposition to both estimators: they have equal squared bias, and $\hat{\theta}$ has no larger variance, so $\text{mse}(\hat{\theta}) \leq \text{mse}(\tilde{\theta})$.

For the equality case, the inequality above is strict unless $\mathbb{E}[\text{Var}(\tilde{\theta} \mid T)] = 0$, that is, unless $\text{Var}(\tilde{\theta} \mid T) = 0$ almost surely, which says exactly that $\tilde{\theta}$ is a function of T . $\square$

So the recipe is: find *any* unbiased estimator, however crude, then condition it on a sufficient statistic. The result is unbiased and at least as good. Let's do this twice.

Example 2.20 (Rao-Blackwellising a Poisson Indicator)

Let $X_1, \dots, X_n$ be i.i.d Poisson(λ), and suppose we wish to estimate $\theta = \mathbb{P}(X_1 = 0) = e^{-\lambda}$. The joint mass function is

$$f_X(x \mid \lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod_i x_i!},$$

so $T = \sum_i X_i$ is sufficient by the factorisation criterion (and note that re-parametrising by $\theta = e^{-\lambda}$ does not change this). Recall $T \sim \text{Poisson}(n\lambda)$.

A very crude unbiased estimator is

$$\tilde{\theta} = \mathbb{1}\{X_1 = 0\},$$

which throws away $X_2, \dots, X_n$ entirely, but does satisfy $\mathbb{E}[\tilde{\theta}] = \mathbb{P}(X_1 = 0) = \theta$. Conditioning on T ,

$$\begin{aligned} \hat{\theta} = \mathbb{E}[\tilde{\theta} \mid T = t] &= \mathbb{P}\left(X_1 = 0 \mid \sum_i X_i = t\right) \\ &= \frac{\mathbb{P}(X_1 = 0) \mathbb{P}\left(\sum_{i \geq 2} X_i = t\right)}{\mathbb{P}\left(\sum_{i \geq 1} X_i = t\right)} \\ &= \frac{e^{-\lambda} \cdot e^{-(n-1)\lambda} ((n-1)\lambda)^t / t!}{e^{-n\lambda} (n\lambda)^t / t!} = \left(\frac{n-1}{n}\right)^t. \end{aligned}$$

So the Rao-Blackwellised estimator is

$$\hat{\theta} = \left(1 - \frac{1}{n}\right)^{\sum_i X_i},$$

which is unbiased and has mean squared error no larger than $\tilde{\theta}$ for every θ . It is also a sensible-looking quantity: writing $\sum_i X_i = n\bar{X}$, we have $\hat{\theta} = (1 - 1/n)^{n\bar{X}} \rightarrow e^{-\bar{X}}$, and $\bar{X} \rightarrow \lambda$ almost surely by the strong law, so $\hat{\theta} \rightarrow e^{-\lambda} = \theta$.

Example 2.21 (Rao-Blackwellising for the Uniform)

Let $X_1, \dots, X_n$ be i.i.d $U[0, \theta]$, and recall $T = \max_i X_i$ is sufficient. Since $\mathbb{E}[X_1] = \theta/2$, the estimator $\hat{\theta} = 2X_1$ is unbiased. Conditioning on $T = t$ and splitting on whether X_1 achieves the maximum,

$$\begin{aligned} \hat{\theta} &= 2 \mathbb{E}\left[X_1 \mid \max_i X_i = t\right] \\ &= 2t \cdot \mathbb{P}\left(X_1 = \max_i X_i \mid \max_i X_i = t\right) \\ &\quad + 2 \mathbb{E}[X_1 \mid X_1 < t] \mathbb{P}\left(X_1 \neq \max_i X_i \mid \max_i X_i = t\right). \end{aligned}$$

By symmetry X_1 is the maximum with probability $1/n$, and given that it is not, X_1 is uniform on $[0, t]$, so has mean $t/2$. Hence

$$\hat{\theta} = \frac{2t}{n} + 2 \cdot \frac{t}{2} \cdot \frac{n-1}{n} = \frac{2t + t(n-1)}{n} = \frac{n+1}{n} \max_i X_i.$$

This is a dramatic improvement. One can compute directly that $\text{Var}(\tilde{\theta}) = \theta^2/3$, while

$$\text{Var}(\hat{\theta}) = \frac{\theta^2}{n(n+2)},$$

so the mean squared error drops from order 1 to order n^{-2} . That is much better

than the usual n^{-1} rate, and reflects the fact that the maximum of a uniform sample converges to θ unusually quickly.

§2.5 Maximum Likelihood Estimation

We now come to the workhorse of the whole subject. The idea is disarmingly simple: having observed x , prefer the value of θ under which what we saw was most probable.

Definition 2.22 (Likelihood)

For fixed observed data x , the **likelihood function** $L : \Theta \rightarrow \mathbb{R}$ is

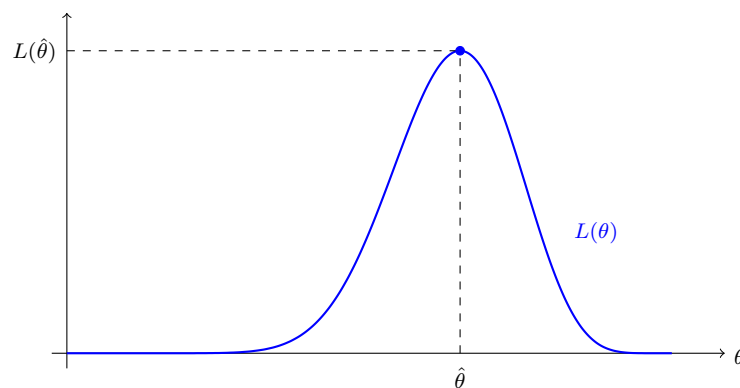
$$L(\theta) = f_X(x \mid \theta) = \prod_{i=1}^n f_{X_i}(x_i \mid \theta),$$

and the **log-likelihood** is $\ell(\theta) = \log L(\theta) = \sum_i \log f_{X_i}(x_i \mid \theta)$.

Note carefully that L is a function of θ , not of x : it is the joint density read ‘sideways’. It is not a probability distribution on Θ and does not integrate to 1.

Definition 2.23 (Maximum Likelihood Estimator)

A **maximum likelihood estimator** (MLE) $\hat{\theta}$ is any value of $\theta \in \Theta$ maximising L , or equivalently (as log is increasing) maximising ℓ .



In well-behaved cases Θ is an open interval and ℓ is differentiable and concave, so we may find $\hat{\theta}$ by solving $\ell'(\theta) = 0$. But this is not a definition, only a technique – and it fails exactly when the likelihood has its maximum on a boundary, which happens more often than one might expect.

Example 2.24 (Bernoulli)

Let $X_1, \dots, X_n$ be i.i.d Bernoulli(p). Then

$$\ell(p) = \sum_i [X_i \log p + (1 - X_i) \log(1 - p)] = \left(\sum_i X_i \right) \log p + \left(n - \sum_i X_i \right) \log(1 - p),$$

so

$$\ell'(p) = \frac{\sum_i X_i}{p} - \frac{n - \sum_i X_i}{1 - p},$$

which vanishes precisely at $p = \bar{X}$. As $\ell'' < 0$ throughout $(0, 1)$ this is the unique maximum, so $\hat{p} = \bar{X}$. This estimator is unbiased.

Example 2.25 (Normal with Both Parameters Unknown)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$. Then

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_i (X_i - \mu)^2.$$

Differentiating,

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (X_i - \mu),$$

which vanishes at $\mu = \bar{X}$ for any σ^2 . Substituting this and differentiating in σ^2 ,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (X_i - \bar{X})^2,$$

which vanishes at $\sigma^2 = S_{XX}/n$. So

$$(\hat{\mu}, \hat{\sigma}^2) = \left(\bar{X}, \frac{S_{XX}}{n} \right).$$

We computed in Section 1 that $\mathbb{E}[S_{XX}] = (n-1)\sigma^2$, so $\hat{\sigma}^2$ is biased downwards, with

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2.$$

The bias vanishes as $n \rightarrow \infty$; we say $\hat{\sigma}^2$ is **asymptotically unbiased**. This behaviour is entirely typical: maximum likelihood estimators of variance parameters are usually a little too small, because they fit the mean to the data first.

Example 2.26 (Uniform on $[0, \theta]$: A Boundary Maximum)

Let $X_1, \dots, X_n$ be i.i.d $U[0, \theta]$. The likelihood is

$$L(\theta) = \frac{1}{\theta^n} \mathbb{1} \left\{ \theta \geq \max_i X_i \right\},$$

which is zero for $\theta < \max_i X_i$ and then strictly decreasing. So it is maximised at

$$\hat{\theta}_{\text{MLE}} = \max_i X_i,$$

and calculus was of no use whatsoever. Comparing with the Rao-Blackwellised estimator $\frac{n+1}{n} \max_i X_i$ obtained above, we see that the MLE is biased downwards, with $\mathbb{E}[\hat{\theta}_{\text{MLE}}] = \frac{n}{n+1} \theta$ – which is obvious in hindsight, since $\max_i X_i < \theta$ always.

Example 2.27 (Poisson)

Let $X_1, \dots, X_n$ be i.i.d $\text{Poisson}(\lambda)$. Then $\ell(\lambda) = -n\lambda + (\sum_i X_i) \log \lambda - \sum_i \log(X_i!)$, and $\ell'(\lambda) = -n + \sum_i X_i/\lambda$ vanishes at $\hat{\lambda} = \bar{X}$. Since $\ell''(\lambda) = -\sum_i X_i/\lambda^2 < 0$ this is the maximum, and it is unbiased.

§2.6 Properties of the Maximum Likelihood Estimator

Maximum likelihood is not just convenient; it has several structural properties that make it the default choice.

Proposition 2.28 (MLEs Depend Only on Sufficient Statistics)

If T is sufficient for θ , then any maximum likelihood estimator is a function of T .

Proof. By the factorisation criterion $L(\theta) = g(T(x), \theta)h(x)$, and $h(x)$ is a positive constant as far as maximisation over θ is concerned. So the maximiser depends on x only through $T(x)$. $\square$

Proposition 2.29 (Invariance)

Let H be a bijection of Θ onto some set Φ , and let $\varphi = H(\theta)$. If $\hat{\theta}$ is a maximum likelihood estimator for θ , then $H(\hat{\theta})$ is a maximum likelihood estimator for φ .

Proof. The likelihood for φ is $L^*(\varphi) = L(H^{-1}(\varphi))$, and as H is a bijection, L^* is maximised at $\varphi = H(\hat{\theta})$ exactly when L is maximised at $\hat{\theta}$. $\square$

This invariance is a genuine convenience: to estimate $e^{-\lambda}$ in the Poisson model we may simply write down $e^{-\bar{X}}$. (Note the contrast with unbiasedness, which has no such invariance – $\mathbb{E}[g(\hat{\theta})] \neq g(\mathbb{E}[\hat{\theta}])$ in general.)

Finally, and most importantly, maximum likelihood estimators are asymptotically normal.

Fact 2.30 (Asymptotic Normality of the MLE)

Under regularity conditions on the model (which we will not state precisely here), the maximum likelihood estimator $\hat{\theta} = \hat{\theta}_n$ based on n i.i.d observations satisfies

$$\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, \Sigma)$$

in distribution as $n \rightarrow \infty$, for a limiting covariance matrix Σ determined by the model. Precisely, for all ‘nice’ sets A ,

$$\mathbb{P}_{\theta}(\sqrt{n}(\hat{\theta} - \theta) \in A) \rightarrow \mathbb{P}(Z \in A), \quad Z \sim N(0, \Sigma).$$

The matrix Σ is the inverse of the *Fisher information* per observation, and one can show that no reasonable estimator can asymptotically beat it – so maximum likelihood is, in a precise sense, asymptotically optimal. We will not prove any of this in this course, but the statement matters: it tells us that for large n the MLE is approximately unbiased,

approximately normal, and has a variance of order $1/n$. That is exactly the input we need to construct approximate confidence intervals, which is where we go next.

Remark. For many models ℓ cannot be maximised in closed form, and $\hat{\theta}$ is found numerically (typically by Newton-Raphson on ℓ' , which converges quickly because ℓ is asymptotically quadratic near its maximum). The asymptotic theory above applies just the same.

§3 Confidence Intervals

A point estimate on its own is not much of an answer. If a poll reports that 52% of voters support a proposal, we want to know whether that number is accurate to within one percentage point or ten. An interval estimate answers this, and confidence intervals are the frequentist way of producing one.

§3.1 Definition and Construction

Definition 3.1 (Confidence Interval)

A $100\gamma\%$ **confidence interval** for a parameter θ is a pair of statistics $A(X) \leq B(X)$ such that

$$\mathbb{P}_{\theta}(A(X) \leq \theta \leq B(X)) = \gamma \quad \text{for all } \theta \in \Theta.$$

The reader should stare at that probability statement for a moment, because everything hinges on what is random in it. The parameter θ is a fixed unknown constant; it is the *endpoints* $A(X)$ and $B(X)$ that are random. We are not saying ‘ θ falls in a fixed interval 95% of the time’; we are saying ‘a random interval catches the fixed θ 95% of the time’. Note also that the requirement is uniform in θ : the coverage must be γ whatever the truth happens to be.

Example 3.2 (Normal Mean, Known Variance)

Let $X_1, \dots, X_n$ be i.i.d $N(\theta, 1)$, and let us find a 95% confidence interval for θ . We know that

$$\bar{X} \sim N\left(\theta, \frac{1}{n}\right), \quad \text{so} \quad Z = \sqrt{n}(\bar{X} - \theta) \sim N(0, 1).$$

The crucial point is that the distribution of Z is *the same for every* θ . So we may pick numbers $a < b$ with $\Phi(b) - \Phi(a) = 0.95$ once and for all, and then

$$0.95 = \mathbb{P}_{\theta}\left(a \leq \sqrt{n}(\bar{X} - \theta) \leq b\right) = \mathbb{P}_{\theta}\left(\bar{X} - \frac{b}{\sqrt{n}} \leq \theta \leq \bar{X} - \frac{a}{\sqrt{n}}\right).$$

Hence $(\bar{X} - b/\sqrt{n}, \bar{X} - a/\sqrt{n})$ is a 95% confidence interval, for *any* such a, b . To make the interval as short as possible we take it symmetric, $-a = b = z_{0.025} \approx 1.96$, where in general $z_{\alpha} = \Phi^{-1}(1 - \alpha)$ denotes the upper α point of $N(0, 1)$. This gives the familiar

$$\bar{X} \pm \frac{1.96}{\sqrt{n}}.$$

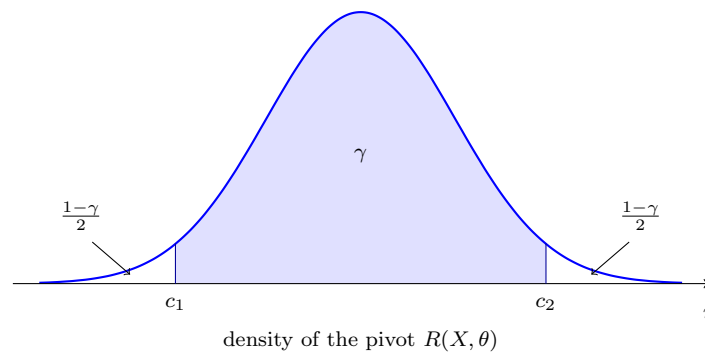
The technique used there deserves a name.

Definition 3.3 (Pivot)

A **pivot** is a function $R(X, \theta)$ of both the data and the parameter whose distribution under $\mathbb{P}_\theta$ is the same for every $\theta \in \Theta$.

A pivot is not a statistic (it involves θ), and its distribution is not a sampling distribution of anything we can compute – but because it does not depend on θ , we can look up its quantiles in advance. The general recipe is then:

- (i) Find a pivot $R(X, \theta)$.
- (ii) Choose constants $c_1 < c_2$ with $\mathbb{P}(c_1 \leq R \leq c_2) = \gamma$, using the (known) distribution of R .
- (iii) Rearrange the inequalities $c_1 \leq R(X, \theta) \leq c_2$ to isolate θ , obtaining $A(X) \leq \theta \leq B(X)$.



Step (iii) is where things can get awkward – the rearrangement need not be possible in closed form, and if R is not monotone in θ the resulting set may not even be an interval. When θ is vector-valued we simply drop the pretence and speak of a **confidence set**: any $A(X) \subseteq \Theta$ with $\mathbb{P}_\theta(\theta \in A(X)) = \gamma$ for all θ .

A useful closure property, immediate from the definition, is the following.

Proposition 3.4

If $(A(X), B(X))$ is a $100\gamma\%$ confidence interval for θ and T is a strictly increasing function, then $(T(A(X)), T(B(X)))$ is a $100\gamma\%$ confidence interval for $T(\theta)$.

Proof. Since T is strictly increasing, $A(X) \leq \theta \leq B(X)$ if and only if $T(A(X)) \leq T(\theta) \leq T(B(X))$, so the two events are identical and have the same probability. $\square$

§3.2 Standard Constructions**Example 3.5 (Normal Variance, Known Mean)**

Let $X_1, \dots, X_n$ be i.i.d $N(0, \sigma^2)$ with σ^2 unknown. Since $X_i/\sigma \sim N(0, 1)$, a natural pivot is

$$R(X, \sigma^2) = \sum_{i=1}^n \frac{X_i^2}{\sigma^2} \sim \chi_n^2,$$

whose distribution is free of σ^2 as required. (We could have used the single observa-

tion X_1/σ , but that would waste $n-1$ of our data points and give a hopelessly wide interval.) Taking $c_1 = \chi_n^2(0.975)$ and $c_2 = \chi_n^2(0.025)$ so that $\mathbb{P}(c_1 \leq R \leq c_2) = 0.95$, and rearranging,

$$0.95 = \mathbb{P}\left(c_1 \leq \frac{\sum_i X_i^2}{\sigma^2} \leq c_2\right) = \mathbb{P}\left(\frac{\sum_i X_i^2}{c_2} \leq \sigma^2 \leq \frac{\sum_i X_i^2}{c_1}\right).$$

Note that the interval is *not* symmetric about any natural estimator, because the χ^2 distribution is skewed.

Example 3.6 (A Bernoulli Proportion)

Let $X_1, \dots, X_n$ be i.i.d Bernoulli(p) with n large. The MLE is $\hat{p} = \bar{X}$, and by the central limit theorem

$$\sqrt{n} \frac{\hat{p} - p}{\sqrt{p(1-p)}} \approx N(0, 1).$$

This is an approximate pivot, but rearranging the resulting inequalities for p requires solving a quadratic (since p appears in the denominator too). The standard dodge is to replace $p(1-p)$ by $\hat{p}(1-\hat{p})$ in the denominator, which is legitimate for large n since $\hat{p} \rightarrow p$. Then

$$\mathbb{P}\left(-z_{0.025} \leq \sqrt{n} \frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})}} \leq z_{0.025}\right) \approx 0.95,$$

giving the approximate 95% confidence interval

$$\hat{p} \pm z_{0.025} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Its width is largest when $\hat{p} = 1/2$, where it is about $2 \times 1.96 \times \frac{1}{2\sqrt{n}} \approx 1/\sqrt{n}$. Replacing $\hat{p}(1-\hat{p})$ by its maximum $1/4$ gives the **conservative** interval $\hat{p} \pm 1/\sqrt{n}$, which has coverage *at least* 95% for every p – this is where the familiar ‘margin of error of 3% on a poll of 1000 people’ comes from.

Example 3.7 (Poisson Rate)

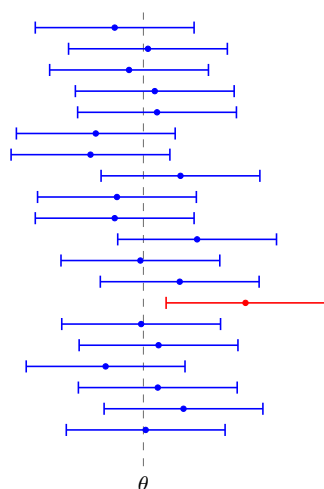
Let $X_1, \dots, X_n$ be i.i.d Poisson(μ) with n large. Since $\text{Var}(X_1) = \mu$, the central limit theorem gives

$$\sqrt{n} \frac{\bar{X} - \mu}{\sqrt{\mu}} \approx N(0, 1),$$

and substituting $\hat{\mu} = \bar{X}$ in the denominator as before yields the approximate 95% interval $\bar{X} \pm 1.96\sqrt{\bar{X}/n}$.

§3.3 Interpreting Confidence Intervals

The frequentist meaning of a confidence interval is a statement about the *procedure*, not about the particular interval we happen to have computed. If we were to repeat the experiment many times, each time constructing the interval by the same recipe, then about $100\gamma\%$ of those intervals would contain the true θ .



Twenty intervals from twenty repetitions of the same experiment are drawn above; one of them (in red) misses the true value. Before the data are seen, the probability that the interval will catch θ is 0.95. After the data are seen, the interval is a specific pair of numbers, and θ either is or is not inside it – there is no residual randomness left to have a probability about.

This is not mere pedantry. Here is an example where the naive reading is plainly absurd.

Example 3.8 (A 50% Interval We Sometimes Know Is Right)

Let X_1, X_2 be i.i.d uniform on $(\theta - \frac{1}{2}, \theta + \frac{1}{2})$. Then

$$\mathbb{P}\left(\min_i X_i \leq \theta \leq \max_i X_i\right) = \mathbb{P}(X_1 \leq \theta \leq X_2) + \mathbb{P}(X_2 \leq \theta \leq X_1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

since each X_i falls above or below θ with probability $1/2$, independently of the other. So the interval $(\min_i X_i, \max_i X_i)$ is a perfectly valid 50% confidence interval.

But now suppose we observe $x_1 = 0.1$ and $x_2 = 0.9$, so $|x_1 - x_2| = 0.8 > 1/2$. Since both observations lie within $1/2$ of θ , we have $\theta > 0.9 - 0.5 = 0.4$ and $\theta < 0.1 + 0.5 = 0.6$, so certainly $\theta \in (0.1, 0.9)$. We are 100% certain, not 50%! Conversely if $x_1 = 0.4$ and $x_2 = 0.45$ we have learned rather little, and 50% feels generous.

The coverage statement is still true – averaged over repetitions, half the intervals contain θ . It is simply not a statement about this particular data set.

If one genuinely wants a probability statement about θ given the data actually observed, one has to be willing to treat θ as a random variable in the first place. That is the Bayesian position, and it is where we turn next.

§4 Bayesian Inference

Everything so far has been **frequentist**: θ is a fixed unknown constant, and all probability statements are about the data, in the context of hypothetical repetitions of the experiment. Bayesian inference takes a different starting point. It treats θ itself as a random variable, with a distribution expressing our uncertainty about it, and then simply conditions on the data.

§4.1 Priors and Posteriors

Definition 4.1 (Prior and Posterior)

The **prior** distribution $\pi(\theta)$ on Θ represents our beliefs about θ before observing any data. Given data X with density (or mass function) $f_X(x | \theta)$, the **posterior** distribution is the conditional distribution of θ given $X = x$, written $\pi(\theta | x)$.

By Bayes' rule,

$$\pi(\theta | x) = \frac{\pi(\theta)f_X(x | \theta)}{f_X(x)}, \quad f_X(x) = \int_{\Theta} f_X(x | \theta)\pi(\theta) d\theta$$

(with the integral replaced by a sum if Θ is discrete). Since the denominator does not involve θ at all, it is nothing but a normalising constant, and in practice we almost always work with the proportionality

$$\pi(\theta | x) \propto \pi(\theta) f_X(x | \theta),$$

in words: *posterior* $\propto$ *prior* $\times$ *likelihood*. This is the entire machinery. The art lies in recognising the right-hand side as a standard distribution up to normalisation, which saves us from ever computing the integral.

Remark. If T is sufficient for θ , the posterior depends on the data only through T . Indeed by the factorisation criterion

$$\pi(\theta | x) \propto \pi(\theta)g(T(x), \theta)h(x) \propto \pi(\theta)g(T(x), \theta),$$

since $h(x)$ is constant in θ . So the Bayesian and frequentist notions of ‘the relevant information in the data’ agree exactly.

Example 4.2 (A Medical Test)

A patient is tested for a disease. Let $\theta = 1$ if they are infected and $\theta = 0$ otherwise, and let $X = 1$ if the test is positive. The likelihood is specified by the **sensitivity** $f_X(1 | 1)$ and the **specificity** $f_X(0 | 0)$ of the test. For the prior we might take $\pi(1)$ to be the prevalence of the disease in the population. Then

$$\pi(1 | X = 1) = \frac{\pi(1)f_X(1 | 1)}{\pi(1)f_X(1 | 1) + \pi(0)f_X(1 | 0)}.$$

If the disease is rare, so $\pi(1) \ll \pi(0)$, this can be small even for an excellent test: with sensitivity and specificity both 99% and prevalence 0.1%, a positive test gives

$$\pi(1 | X = 1) = \frac{0.001 \times 0.99}{0.001 \times 0.99 + 0.999 \times 0.01} \approx 0.09.$$

The false positives from the huge healthy population swamp the true positives. This is the classical illustration of why the prior cannot be ignored.

§4.2 Conjugate Priors

The computation is especially pleasant when prior and posterior belong to the same family.

Definition 4.3 (Conjugacy)

A family of priors is **conjugate** to a likelihood if the posterior always lies in the same family.

Example 4.4 (Beta-Binomial)

Let θ be the mortality rate for a particular operation, and model the number of deaths in n operations as $X \sim \text{Bin}(n, \theta)$. Suppose data from other hospitals suggest the rate lies between 3% and 20% with an average around 10%. A natural family of priors on $[0, 1]$ is the beta family,

$$\pi(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}, \quad \theta \sim \text{Beta}(a, b),$$

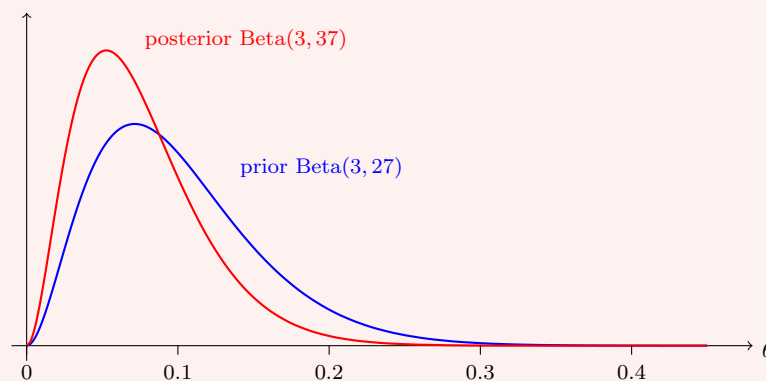
which has mean $a/(a+b)$. Taking $a = 3$, $b = 27$ gives $\mathbb{E}[\theta] = 0.1$ and $\mathbb{P}(0.03 < \theta < 0.2) \approx 0.9$, matching our beliefs well.

Now suppose we observe x deaths in n operations. Then

$$\pi(\theta | x) \propto \theta^{a-1}(1-\theta)^{b-1} \cdot \theta^x(1-\theta)^{n-x} = \theta^{x+a-1}(1-\theta)^{n-x+b-1},$$

which we recognise immediately as $\text{Beta}(a+x, b+n-x)$. So the beta family is conjugate to the binomial likelihood, and the update rule is beautifully simple: add the successes to a , add the failures to b .

With $n = 10$ operations and $x = 0$ deaths, the posterior is $\text{Beta}(3, 37)$.



The posterior has both a smaller mean ($3/40 = 0.075$ against $3/30 = 0.1$) and a smaller variance than the prior: observing no deaths has shifted our beliefs downwards and sharpened them. Notice also how modest the shift is – ten operations is not much data against a prior worth 30 pseudo-observations, and the Bayesian machinery weights them accordingly.

Example 4.5 (Gamma-Poisson)

Let $X_1, \dots, X_n$ be i.i.d $\text{Poisson}(\theta)$ with prior $\theta \sim \text{Gamma}(a, b)$, so $\pi(\theta) \propto \theta^{a-1}e^{-b\theta}$. Then

$$\pi(\theta | x) \propto \theta^{a-1}e^{-b\theta} \cdot \theta^{\sum x_i} e^{-n\theta} = \theta^{a+\sum x_i-1} e^{-(b+n)\theta},$$

so the posterior is $\text{Gamma}(a + \sum_i x_i, b + n)$: the gamma family is conjugate to the Poisson.

Example 4.6 (Normal-Normal)

Let $X_1, \dots, X_n$ be i.i.d $N(\theta, \sigma^2)$ with σ^2 known, and give θ the prior $N(\mu_0, \tau_0^2)$. Then

$$\pi(\theta | x) \propto \exp\left(-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\right) \exp\left(-\frac{\sum_i (x_i - \theta)^2}{2\sigma^2}\right).$$

The exponent is a quadratic in θ , so the posterior is normal, and completing the square gives $N(\mu_1, \tau_1^2)$ with

$$\frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad \mu_1 = \tau_1^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right).$$

The interpretation is very clean if we think in terms of **precisions** (reciprocal variances): precisions add, and the posterior mean is the precision-weighted average of the prior mean and the sample mean. As $n \rightarrow \infty$ the data term dominates and $\mu_1 \rightarrow \bar{x}$, so the prior is eventually washed out – which is reassuring.

§4.3 Point Estimates as Optimal Decisions

The posterior is the complete Bayesian answer. But often we must actually *do* something – commit to a single number, or a single course of action. Decision theory tells us how.

We choose a decision δ from a set Δ of available actions, and specify a **loss function** $L(\theta, \delta)$ giving the penalty incurred for taking action δ when the truth is θ . Having observed x , we then choose δ to minimise the **posterior expected loss**

$$h(\delta) = \int_{\Theta} L(\theta, \delta) \pi(\theta | x) d\theta.$$

Definition 4.7 (Bayes Estimator)

When $\Delta = \Theta$ (so that our ‘action’ is to report an estimate of θ), the minimiser $\hat{\theta}^{(B)}$ of $h(\delta)$ is the **Bayes estimator** of θ for the loss L .

Different losses give different estimators, and pleasingly the three most natural losses give the three most natural summaries of the posterior.

Example 4.8 (Quadratic Loss Gives the Posterior Mean)

Take $L(\theta, \delta) = (\theta - \delta)^2$. Then

$$h(\delta) = \int_{\Theta} (\theta - \delta)^2 \pi(\theta | x) d\theta,$$

and differentiating under the integral,

$$h'(\delta) = -2 \int_{\Theta} (\theta - \delta) \pi(\theta | x) d\theta,$$

which vanishes when $\delta = \int_{\Theta} \theta \pi(\theta | x) d\theta$. As $h''(\delta) = 2 > 0$ this is the minimum. So the Bayes estimator under quadratic loss is the *posterior mean*.

Example 4.9 (Absolute Loss Gives the Posterior Median)

Take $L(\theta, \delta) = |\theta - \delta|$. Splitting the integral at δ ,

$$h(\delta) = \int_{-\infty}^{\delta} (\delta - \theta) \pi(\theta | x) d\theta + \int_{\delta}^{\infty} (\theta - \delta) \pi(\theta | x) d\theta,$$

and differentiating (the boundary terms cancel since the integrands vanish at $\theta = \delta$),

$$h'(\delta) = \int_{-\infty}^{\delta} \pi(\theta | x) d\theta - \int_{\delta}^{\infty} \pi(\theta | x) d\theta.$$

This vanishes when the two tails have equal mass, i.e. at the *posterior median*. Since h' is increasing, this is the minimum.

Example 4.10 (Zero-One Loss Gives the Posterior Mode)

For $\varepsilon > 0$ take $L(\theta, \delta) = \mathbb{1}\{|\theta - \delta| > \varepsilon\}$, which penalises us only if we miss by more than ε . Then

$$h(\delta) = 1 - \int_{\delta-\varepsilon}^{\delta+\varepsilon} \pi(\theta | x) d\theta,$$

which is minimised by centring the window on the region of highest posterior density. As $\varepsilon \rightarrow 0$ this converges to the *posterior mode*, also called the maximum a posteriori (MAP) estimate. Note that with a flat prior the posterior is proportional to the likelihood, so the MAP estimate is then exactly the maximum likelihood estimate – a satisfying point of contact between the two philosophies.

Example 4.11 (Beta-Binomial, Continued)

In the surgery example the posterior was $\text{Beta}(3, 37)$. Under quadratic loss the Bayes estimator is the posterior mean $3/40 = 0.075$, while the posterior mode is $(3 - 1)/(3 + 37 - 2) = 2/38 \approx 0.053$. Compare the maximum likelihood estimate $\hat{\theta} = x/n = 0$, which asserts that the operation is perfectly safe on the basis of ten cases. The prior is doing real work here, and doing it sensibly.

§4.4 Credible Intervals**Definition 4.12** (Credible Interval)

A $100\gamma\%$ **credible interval** for θ is an interval $(A(x), B(x))$ with

$$\mathbb{P}(A(x) \leq \theta \leq B(x) | X = x) = \int_{A(x)}^{B(x)} \pi(\theta | x) d\theta = \gamma.$$

Superficially this looks identical to a confidence interval, but the probability means something quite different – and much closer to what people usually want. Here θ is the random quantity and the data are fixed, so we really may say: ‘given the data I have, there is probability γ that θ lies in this interval’. That is precisely the statement that was illegitimate for confidence intervals.

The price, of course, is that the answer depends on the prior. Two statisticians with

different priors will report different credible intervals from the same data, and neither is wrong. Note also that credible intervals are not unique: any A, B capturing γ of the posterior mass will do. Common choices are the **equal-tailed** interval, running between the $\frac{1-\gamma}{2}$ and $\frac{1+\gamma}{2}$ posterior quantiles, and the **highest posterior density** interval, which is the shortest such interval and consists of the points with the largest posterior density.

Example 4.13

Continuing the surgery example, the equal-tailed 95% credible interval for the mortality rate under the $\text{Beta}(3, 37)$ posterior runs between the 2.5% and 97.5% points of that distribution, which are approximately 0.016 and 0.173. So having seen ten operations with no deaths, we would report a mortality rate of about 7.5%, with a credible interval of roughly (1.6%, 17.3%).

§5 Hypothesis Testing

Estimation asks ‘what is θ ?’. Hypothesis testing asks a cruder but often more useful question: ‘is there evidence in the data against a particular claim about θ ?’. This is the form most scientific questions actually take – does the drug work, is the coin fair, does the data fit the model – and the framework we build here is the one used, for better or worse, throughout the empirical sciences.

§5.1 Hypotheses, Tests and Errors

Definition 5.1 (Hypothesis)

A **hypothesis** is an assertion about the distribution of the data X . The **null hypothesis** H_0 is the default or baseline assertion; the **alternative hypothesis** H_1 is what we entertain if H_0 is found wanting.

A hypothesis is **simple** if it completely specifies the distribution of X , and **composite** otherwise.

Example 5.2

Let $X_1, \dots, X_n$ be i.i.d $\text{Bernoulli}(\theta)$. We might test $H_0 : \theta = \frac{1}{2}$ against $H_1 : \theta = \frac{3}{4}$, in which case both hypotheses are simple; or $H_0 : \theta = \frac{1}{2}$ against $H_1 : \theta \neq \frac{1}{2}$, in which case H_1 is composite.

Example 5.3

Suppose the X_i take values in $\{0, 1, 2, \dots\}$ and we test $H_0 : X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$ for *some* λ , against H_1 : the X_i have some other distribution. Here H_0 is composite (because λ is unspecified), and the test is a **goodness-of-fit** test. We will return to these later in this section.

Definition 5.4 (Test)

A **test** of H_0 is specified by a **critical region** (or rejection region) $C \subseteq \mathcal{X}$, where

$\mathcal{X}$ is the sample space. If $X \in C$ we **reject** H_0 ; otherwise we **fail to reject** H_0 . The complement $A = \mathcal{X} \setminus C$ is the **acceptance region**.

The asymmetric language is deliberate. Failing to reject H_0 is not evidence *for* H_0 – it may simply mean we had too little data to detect a departure. There are two ways to go wrong:

Definition 5.5 (Type I and Type II Errors)

A **type I error** (a false positive) is rejecting H_0 when it is true; a **type II error** (a false negative) is failing to reject H_0 when it is false. When H_0 and H_1 are simple we write

$$\alpha = \mathbb{P}_{H_0}(X \in C), \quad \beta = \mathbb{P}_{H_1}(X \notin C).$$

We call α the **size** of the test and $1 - \beta = \mathbb{P}_{H_1}(X \in C)$ its **power**.

The two error probabilities trade off against each other: shrinking C reduces α and increases β . The classical resolution of this tension is to fix an acceptable type I error rate α (conventionally 0.05 or 0.01) and then, subject to that constraint, make the power as large as possible. Note the asymmetry this builds in: H_0 is protected, and we reject it only on strong evidence. Note also that computing α requires only H_0 , whereas computing β requires us to specify H_1 precisely – which is one reason simple alternatives are so much easier to handle.

§5.2 The Neyman-Pearson Lemma

Suppose both hypotheses are simple, with X having density (or mass function) f_i under H_i . Which critical region should we choose?

Definition 5.6 (Likelihood Ratio Test)

The **likelihood ratio statistic** is

$$\Lambda_x(H_0; H_1) = \frac{f_1(x)}{f_0(x)},$$

and the **likelihood ratio test** has critical region $C = \{x : \Lambda_x(H_0; H_1) > k\}$ for some constant k .

In other words – reject H_0 exactly when the data are sufficiently more probable under H_1 than under H_0 . This is an obviously reasonable thing to do, and the remarkable content of the next result is that it is *optimal*.

Lemma 5.7 (Neyman-Pearson)

Let H_0 and H_1 be simple hypotheses with densities f_0, f_1 that are positive on the same set. Suppose that for some $k > 0$ the likelihood ratio test with critical region $C = \{x : f_1(x)/f_0(x) > k\}$ has size α . Then among all tests of size at most α , this test has the largest power.

Proof. Write $\overline{C}$ for the complement of C , and let

$$\alpha = \int_C f_0(x) \, dx, \quad \beta = \int_{\overline{C}} f_1(x) \, dx$$

be the error probabilities of the likelihood ratio test. Let C^* be the critical region of any other test, with error probabilities α^*, β^* , and suppose $\alpha^* \leq \alpha$. We must show $\beta \leq \beta^*$.

Cancelling the contribution from $\overline{C} \cap \overline{C}^*$, which is common to both,

$$\beta - \beta^* = \int_{\overline{C}} f_1 - \int_{\overline{C}^*} f_1 = \int_{\overline{C} \cap C^*} f_1 - \int_{C \cap \overline{C}^*} f_1.$$

Now on $\overline{C}$ we have $f_1 \leq k f_0$, and on C we have $f_1 \geq k f_0$. Applying these to the two integrals respectively (the first is over a subset of $\overline{C}$, the second over a subset of C),

$$\begin{aligned} \beta - \beta^* &\leq k \int_{\overline{C} \cap C^*} f_0 - k \int_{C \cap \overline{C}^*} f_0 \\ &= k \left[\int_{\overline{C} \cap C^*} f_0 + \int_{C \cap C^*} f_0 - \int_{C \cap C^*} f_0 - \int_{C \cap \overline{C}^*} f_0 \right] \\ &= k \left[\int_{C^*} f_0 - \int_C f_0 \right] = k [\alpha^* - \alpha] \leq 0, \end{aligned}$$

where in the middle line we added and subtracted the same quantity in order to reassemble the integrals over C^* and over C . Hence $\beta \leq \beta^*$, and the likelihood ratio test has power $1 - \beta \geq 1 - \beta^*$. $\square$

Remark. The hypothesis that a likelihood ratio test of *exactly* size α exists is a real restriction, and typically fails for discrete data, where the achievable sizes form a finite set. The fix is to allow **randomised tests**, which for certain boundary values of x reject H_0 with some probability strictly between 0 and 1; every size α is then achievable, and the lemma holds in that generality.

Example 5.8 (The Z -Test)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known, and test $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1$ where $\mu_1 > \mu_0$. The likelihood ratio is

$$\begin{aligned} \Lambda_x &= \frac{(2\pi\sigma_0^2)^{-n/2} \exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \mu_1)^2\right)}{(2\pi\sigma_0^2)^{-n/2} \exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \mu_0)^2\right)} \\ &= \exp\left(\underbrace{\frac{\mu_1 - \mu_0}{\sigma_0^2}}_{>0} n\bar{x} + \frac{n(\mu_0^2 - \mu_1^2)}{2\sigma_0^2}\right), \end{aligned}$$

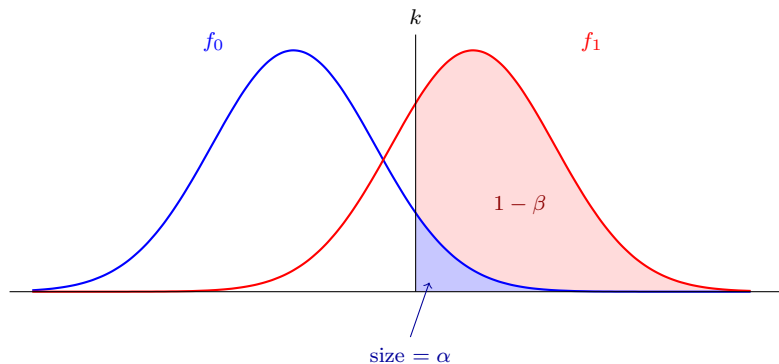
which depends on the data only through $\bar{x}$, and is strictly increasing in it. Hence

$\Lambda_x > k$ if and only if $\bar{x} > k'$, if and only if

$$Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > k''$$

for suitable constants. Under H_0 we have $Z \sim N(0, 1)$, so to achieve size α we take $k'' = z_\alpha = \Phi^{-1}(1 - \alpha)$. This is the **Z-test**, and by Neyman-Pearson it is the most powerful test of its size.

The geometry of what is going on is worth a picture. Below are the two densities of the test statistic Z under H_0 and H_1 , with the critical region $\{Z > k\}$ shaded under each.



Moving k to the right shrinks the blue shaded area (the size) but also shrinks the red one (the power). The Neyman-Pearson lemma says that among all ways of carving a region of blue area α out of the line, taking a right-hand tail of the likelihood ratio captures the most red.

§5.3 p -Values

Reporting only ‘reject at the 5% level’ throws away information: it does not distinguish a marginal result from an overwhelming one. The p -value records how extreme the observed data actually were.

Definition 5.9 (p -Value)

Suppose the critical region has the form $C = \{x : T(x) > k\}$ for a test statistic T , and let x^* be the observed data. The **p -value** is

$$p = \mathbb{P}_{H_0}(T(X) \geq T(x^*)).$$

Equivalently, the p -value is the smallest size α at which we would have rejected H_0 with these data. Small p means the observed data were surprising under H_0 .

Example 5.10

Continuing the Z -test with $\mu_0 = 5$, $\sigma_0 = 1$, $\alpha = 0.05$, suppose we observe $x^* = (5.1, 5.5, 4.9, 5.3)$. Then $\bar{x}^* = 5.2$ and $z^* = \sqrt{4}(5.2 - 5)/1 = 0.4$. The critical region is $\{Z > 1.645\}$, so we do not reject H_0 ; the p -value is $1 - \Phi(0.4) \approx 0.34$, so a value at least this extreme would arise about a third of the time even if H_0 were true.

Proposition 5.11

If the test statistic T has a continuous distribution, then under H_0 the p -value is uniformly distributed on $[0, 1]$.

Proof. Let F be the distribution function of T under H_0 , so $p = 1 - F(T)$. Then for $u \in (0, 1)$,

$$\begin{aligned}\mathbb{P}_{H_0}(p \leq u) &= \mathbb{P}_{H_0}(1 - F(T) \leq u) = \mathbb{P}_{H_0}(F(T) \geq 1 - u) \\ &= \mathbb{P}_{H_0}(T \geq F^{-1}(1 - u)) = 1 - F(F^{-1}(1 - u)) = u. \square\end{aligned}$$

This is a useful sanity check, and also a warning: under the null, p -values below 0.05 occur exactly 5% of the time by construction. If one tests twenty independent null hypotheses, one expects a ‘significant’ result by chance alone.

§5.4 Composite Hypotheses and Power Functions

Now let $X \sim f_X(\cdot | \theta)$ with $\theta \in \Theta$, and consider $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$, where Θ_0, Θ_1 are disjoint. The error probabilities now depend on which particular θ obtains, so we need a function rather than a number.

Definition 5.12 (Power Function, Size, UMP)

The **power function** of a test with critical region C is

$$W(\theta) = \mathbb{P}_\theta(X \in C), \quad \theta \in \Theta.$$

Its **size** is $\alpha = \sup_{\theta \in \Theta_0} W(\theta)$, the worst-case type I error rate. A test with power function W and size at most α is **uniformly most powerful** (UMP) of size α if for every other test with power function W^* and size at most α ,

$$W(\theta) \geq W^*(\theta) \quad \text{for all } \theta \in \Theta_1.$$

The word ‘uniformly’ is doing the work: we are asking for a single test that is simultaneously best against every alternative. Such tests need not exist – and typically do not, for two-sided alternatives, since the best test against $\theta > \theta_0$ points in the wrong direction for $\theta < \theta_0$. For one-sided alternatives in simple models, however, they often do.

Example 5.13 (One-Sided Test for a Normal Mean)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known, and consider

$$H_0 : \mu \leq \mu_0, \quad H_1 : \mu > \mu_0.$$

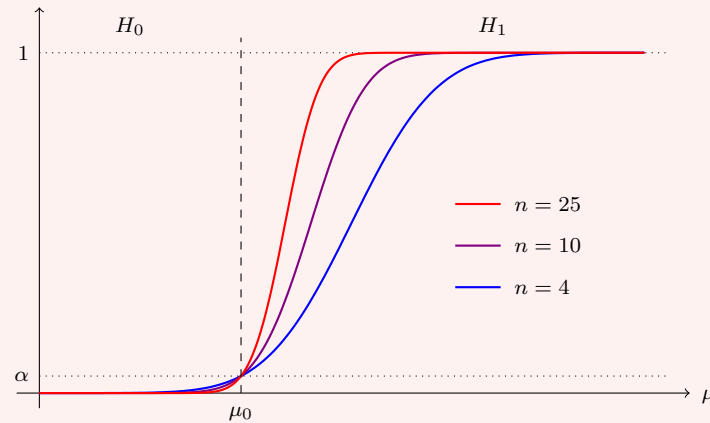
We claim the Z -test with $C = \{Z > z_\alpha\}$, where $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma_0$, is UMP of size α .

First we compute the power function. Under $\mathbb{P}_\mu$ we have $\sqrt{n}(\bar{X} - \mu)/\sigma_0 \sim N(0, 1)$, so

$$W(\mu) = \mathbb{P}_\mu \left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > z_\alpha \right)$$

$$\begin{aligned}
&= \mathbb{P}_\mu \left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma_0} > z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma_0} \right) \\
&= 1 - \Phi \left(z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma_0} \right).
\end{aligned}$$

This is strictly increasing in μ , so $\sup_{\mu \leq \mu_0} W(\mu) = W(\mu_0) = 1 - \Phi(z_\alpha) = \alpha$, and the test has size α as claimed.



The picture shows what a good power function looks like: it passes through α at the boundary μ_0 , stays below α on Θ_0 , and climbs towards 1 on Θ_1 , the more steeply the more data we have.

Now for optimality. The key observation is that the critical region C does not depend on μ_1 at all – it was determined by μ_0 and α alone. So fix any $\mu_1 > \mu_0$ and consider the simple hypotheses $H'_0 : \mu = \mu_0$ and $H'_1 : \mu = \mu_1$. By the previous example, C is precisely the likelihood ratio critical region for H'_0 against H'_1 , of size α .

Let C^* be any test of H_0 against H_1 with size at most α . Regarded as a test of H'_0 against H'_1 , its size is

$$W^*(\mu_0) \leq \sup_{\mu \leq \mu_0} W^*(\mu) \leq \alpha.$$

So by the Neyman-Pearson lemma, $W(\mu_1) \geq W^*(\mu_1)$. Since $\mu_1 > \mu_0$ was arbitrary, the test is UMP.

§5.5 Generalised Likelihood Ratio Tests

For genuinely composite hypotheses there is no likelihood ratio to speak of, since neither hypothesis specifies a single density. The natural fix is to compare the best each hypothesis can do.

Definition 5.14 (Generalised Likelihood Ratio)

Suppose the hypotheses are **nested**, $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \in \Theta_1$ with $\Theta_0 \subseteq \Theta_1$. The **generalised likelihood ratio** is

$$\Lambda_x(H_0; H_1) = \frac{\sup_{\theta \in \Theta_1} f_X(x | \theta)}{\sup_{\theta \in \Theta_0} f_X(x | \theta)},$$

and the **generalised likelihood ratio test** rejects H_0 when Λ_x is large.

Since $\Theta_0 \subseteq \Theta_1$ we always have $\Lambda_x \geq 1$: the larger model can always fit at least as well. The question is whether it fits *enough* better to justify its extra parameters, and the test statistic $2 \log \Lambda_x$ measures exactly this.

Example 5.15 (Two-Sided Test for a Normal Mean)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known, and test $H_0 : \mu = \mu_0$ against $H_1 : \mu \in \mathbb{R}$. The numerator is maximised at the MLE $\hat{\mu} = \bar{X}$, so

$$\Lambda_x = \frac{\exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \bar{x})^2\right)}{\exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \mu_0)^2\right)}.$$

Expanding $\sum_i (x_i - \mu_0)^2 = \sum_i (x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2$, everything cancels except

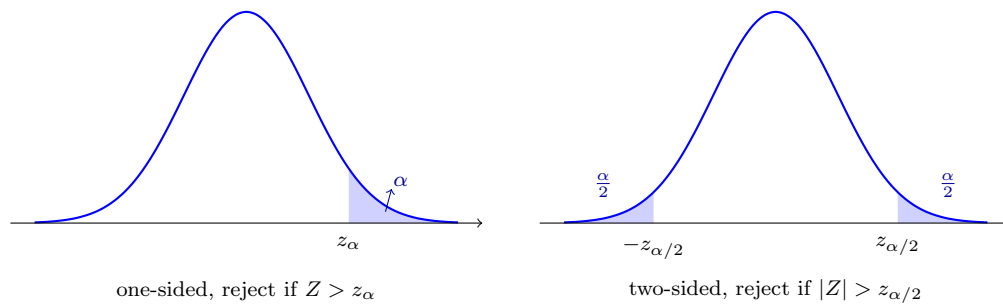
$$2 \log \Lambda_x = \frac{n(\bar{x} - \mu_0)^2}{\sigma_0^2} = \left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} \right)^2.$$

Under H_0 the bracket is $N(0, 1)$, so $2 \log \Lambda_x \sim \chi_1^2$ *exactly*, and the size- α test rejects when

$$\frac{n(\bar{X} - \mu_0)^2}{\sigma_0^2} > \chi_1^2(\alpha), \quad \text{equivalently} \quad \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} \right| > z_{\alpha/2}.$$

This is a **two-sided** test: the critical region consists of both tails in terms of the original statistic.

It is worth seeing the one- and two-sided critical regions side by side, since choosing between them is a modelling decision that must be made *before* looking at the data.



In the example above the exact distribution of $2 \log \Lambda$ was available. Usually it is not, and we appeal to the following remarkable theorem, whose content is that the χ_1^2 we just found was no accident.

Definition 5.16 (Dimension of a Hypothesis)

The **dimension** of $H_0 : \theta \in \Theta_0$ is the number of free parameters in Θ_0 .

Example 5.17

If $\Theta_0 = \{\theta \in \mathbb{R}^k : \theta_1 = \dots = \theta_p = 0\}$ then $\dim \Theta_0 = k - p$. More generally if $A \in \mathbb{R}^{p \times k}$ has linearly independent rows and $\Theta_0 = \{\theta \in \mathbb{R}^k : A\theta = b\}$, then $\dim \Theta_0 = k - p$: each independent linear constraint costs one dimension.

Theorem 5.18 (Wilks)

Suppose $\Theta_0 \subseteq \Theta_1$ with $\dim \Theta_1 - \dim \Theta_0 = p$, and let $X = (X_1, \dots, X_n)$ be i.i.d from $f_X(\cdot | \theta)$ with the true θ in the interior of Θ_0 . Then under regularity conditions, as $n \rightarrow \infty$,

$$2 \log \Lambda_X \rightarrow \chi_p^2$$

in distribution; that is, $\mathbb{P}_\theta(2 \log \Lambda_X \leq \ell) \rightarrow \mathbb{P}(\Xi \leq \ell)$ for every $\ell > 0$, where $\Xi \sim \chi_p^2$.

We will not prove this (the proof is a Taylor expansion of ℓ about the MLE, combined with asymptotic normality of the MLE). But it is enormously useful: it says that whatever the model, the price of p extra parameters is measured on the same universal χ_p^2 scale. In particular the size- α generalised likelihood ratio test rejects when $2 \log \Lambda_X > \chi_p^2(\alpha)$, and this has size approximately α for large n .

Example 5.19

In the two-sided normal example, $\dim \Theta_1 = 1$ and $\dim \Theta_0 = 0$, so $p = 1$ and Wilks' theorem predicts $2 \log \Lambda \approx \chi_1^2$ – which we verified exactly, for every n .

§5.6 Goodness of Fit

We now apply this machinery to the question of whether a model fits at all. Let $X_1, \dots, X_n$ be i.i.d taking values in $\{1, \dots, k\}$, and write $p_i = \mathbb{P}(X_1 = i)$ and

$$N_i = \#\{j : X_j = i\},$$

so $\sum_i p_i = 1$ and $\sum_i N_i = n$. Then $(N_1, \dots, N_k) \sim \text{Multinomial}(n; p_1, \dots, p_k)$, and the parameter space $\Theta_1 = \{p : p_i \geq 0, \sum_i p_i = 1\}$ has dimension $k - 1$.

The **goodness-of-fit** problem is to test $H_0 : p = \tilde{p}$ for a specified vector $\tilde{p}$, against the unrestricted H_1 . The likelihood is

$$L(p) \propto p_1^{N_1} \cdots p_k^{N_k}, \quad \ell(p) = \text{const} + \sum_i N_i \log p_i.$$

Lemma 5.20

The maximum likelihood estimator of p over Θ_1 is $\hat{p}_i = N_i/n$.

Proof. Maximise ℓ subject to $\sum_i p_i = 1$ by Lagrange multipliers: setting $\mathcal{L} = \sum_i N_i \log p_i - \lambda(\sum_i p_i - 1)$ and differentiating gives $N_i/p_i = \lambda$, so $p_i = N_i/\lambda$. Summing over i and using the constraint gives $\lambda = n$. $\square$

So the generalised likelihood ratio statistic is

$$2 \log \Lambda = 2(\ell(\hat{p}) - \ell(\tilde{p})) = 2 \sum_{i=1}^k N_i \log \left(\frac{N_i}{n\tilde{p}_i} \right) = 2 \sum_{i=1}^k o_i \log \frac{o_i}{e_i},$$

writing $o_i = N_i$ for the **observed** counts and $e_i = n\tilde{p}_i$ for the counts **expected** under H_0 . Since $\dim \Theta_1 - \dim \Theta_0 = (k - 1) - 0 = k - 1$, Wilks' theorem tells us to refer this to χ_{k-1}^2 .

§5.7 Pearson's Chi-Squared Statistic

There is a classical approximation to $2 \log \Lambda$ which is easier to compute by hand and is what is usually reported.

Proposition 5.21 (Pearson's Statistic)

With $\delta_i = o_i - e_i$, and assuming each δ_i/e_i is small,

$$2 \log \Lambda \approx \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}.$$

Proof. Write $o_i = e_i + \delta_i$, so

$$2 \log \Lambda = 2 \sum_i (e_i + \delta_i) \log \left(1 + \frac{\delta_i}{e_i} \right).$$

Expanding $\log(1 + u) = u - u^2/2 + O(u^3)$ with $u = \delta_i/e_i$,

$$2 \log \Lambda \approx 2 \sum_i (e_i + \delta_i) \left(\frac{\delta_i}{e_i} - \frac{\delta_i^2}{2e_i^2} \right) = 2 \sum_i \left(\delta_i + \frac{\delta_i^2}{e_i} - \frac{\delta_i^2}{2e_i} \right) + O \left(\sum_i \frac{\delta_i^3}{e_i^2} \right).$$

Now $\sum_i \delta_i = \sum_i o_i - \sum_i e_i = n - n = 0$, so the first term vanishes and we are left with $\sum_i \delta_i^2/e_i$. $\square$

This is **Pearson's χ^2 statistic**, and it is referred to the same χ_{k-1}^2 distribution. The rule of thumb is that the approximation (both to $2 \log \Lambda$ and of both to χ^2) is adequate provided every $e_i \geq 5$.

Example 5.22 (Mendel's Peas)

Mendel crossed pea plants and classified 556 offspring by shape and colour, obtaining

$$N = (\underbrace{315}_{\text{smooth green}}, \underbrace{108}_{\text{smooth yellow}}, \underbrace{102}_{\text{wrinkled green}}, \underbrace{31}_{\text{wrinkled yellow}}).$$

His theory of independent assortment of two dominant/recessive traits predicts

$$H_0 : \tilde{p} = \left(\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16} \right),$$

giving expected counts $e = (312.75, 104.25, 104.25, 34.75)$. Computing,

$$2 \log \Lambda = 0.618, \quad \sum_i \frac{(o_i - e_i)^2}{e_i} = 0.604,$$

which we refer to χ_3^2 . Since $\chi_3^2(0.05) = 7.815$, we do not reject H_0 at the 5% level. Indeed the p -value is $\mathbb{P}(\chi_3^2 > 0.60) \approx 0.90$: the fit is not merely acceptable, it is suspiciously good.

The null hypothesis need not fully specify $\tilde{p}$; often it says only that p lies in some smaller family.

Example 5.23 (Composite Null: Hardy-Weinberg)

Individuals have one of three genotypes with probabilities

$$(p_1, p_2, p_3) = (\theta^2, 2\theta(1 - \theta), (1 - \theta)^2)$$

for some allele frequency $\theta \in (0, 1)$, so $\dim \Theta_0 = 1$. The log-likelihood under H_0 is

$$\ell(\theta) = \text{const} + (2N_1 + N_2) \log \theta + (2N_3 + N_2) \log(1 - \theta),$$

which is maximised at

$$\hat{\theta} = \frac{2N_1 + N_2}{2n}$$

– the observed frequency of the allele, as one would hope. Then

$$2 \log \Lambda = 2 \left(\ell(\hat{p}) - \ell(\hat{\theta}) \right) = 2 \sum_i o_i \log \frac{o_i}{e_i} \approx \sum_i \frac{(o_i - e_i)^2}{e_i}, \quad e_i = n p_i(\hat{\theta}),$$

which is now referred to χ_p^2 with $p = \dim \Theta_1 - \dim \Theta_0 = 2 - 1 = 1$. The general rule: *each parameter estimated under the null costs one degree of freedom.*

§5.8 Contingency Tables

Two of the most common applications of the χ^2 test involve two-way tables of counts, and it is a pleasant surprise that they turn out to be the same test.

§5.8.1 Testing Independence

Suppose we observe i.i.d pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ where X_ℓ takes values in $\{1, \dots, r\}$ and Y_ℓ in $\{1, \dots, c\}$, and we wish to test whether X and Y are independent. The data are summarised by the **contingency table**

$$N_{ij} = \# \{ \ell : (X_\ell, Y_\ell) = (i, j) \},$$

and (N_{ij}) , flattened into a vector, is multinomial with probabilities (p_{ij}) . Writing $p_{i+} = \sum_j p_{ij}$ and $p_{+j} = \sum_i p_{ij}$ for the marginals, the hypotheses are

$$H_0 : p_{ij} = p_{i+}p_{+j} \text{ for all } i, j, \quad H_1 : p \text{ unrestricted.}$$

The unrestricted MLE is $\hat{p}_{ij} = N_{ij}/n$ as before. Under H_0 the free parameters are the two marginal vectors, each constrained to sum to 1, and a Lagrange multiplier computation identical to the one above gives

$$\hat{p}_{i+} = \frac{N_{i+}}{n}, \quad \hat{p}_{+j} = \frac{N_{+j}}{n}.$$

Hence, with $o_{ij} = N_{ij}$ and $e_{ij} = n\hat{p}_{i+}\hat{p}_{+j} = N_{i+}N_{+j}/n$,

$$2 \log \Lambda = 2 \sum_{i,j} N_{ij} \log \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} = 2 \sum_{i,j} o_{ij} \log \frac{o_{ij}}{e_{ij}} \approx \sum_{i,j} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}.$$

The degrees of freedom are

$$p = \dim \Theta_1 - \dim \Theta_0 = (rc - 1) - [(r - 1) + (c - 1)] = (r - 1)(c - 1),$$

so we refer the statistic to $\chi_{(r-1)(c-1)}^2$.

§5.8.2 Testing Homogeneity

Now suppose the row totals are fixed by design rather than random – for instance, we allocate a fixed number of patients to each of several treatments. Then the rows are independent multinomials,

$$(N_{i1}, \dots, N_{ic}) \sim \text{Multinomial}(n_{i+}; p_{i1}, \dots, p_{ic}),$$

and the natural question is whether the treatments differ:

$$H_0 : p_{1j} = p_{2j} = \dots = p_{rj} \text{ for all } j, \quad H_1 : \text{each row an arbitrary probability vector.}$$

The log-likelihood is $\ell(p) = \text{const} + \sum_{i,j} N_{ij} \log p_{ij}$, exactly as in the independence problem. Maximising row by row gives $\hat{p}_{ij} = N_{ij}/n_{i+}$ under H_1 ; under H_0 , writing p_j for the common value, $\ell = \text{const} + \sum_j N_{+j} \log p_j$ is maximised at $\hat{p}_j = N_{+j}/n_{++}$. So

$$2 \log \Lambda = 2 \sum_{i,j} N_{ij} \log \frac{N_{ij}/n_{i+}}{N_{+j}/n_{++}} = 2 \sum_{i,j} o_{ij} \log \frac{o_{ij}}{e_{ij}}, \quad e_{ij} = \frac{n_{i+}N_{+j}}{n_{++}},$$

which is *identical* to the statistic for independence, with identical expected counts. The degrees of freedom also agree:

$$p = r(c-1) - (c-1) = (r-1)(c-1).$$

So although the sampling schemes and the hypotheses are conceptually quite different, the two tests are operationally the same: we reject one exactly when we reject the other.

Example 5.24 (A Clinical Trial)

150 patients are randomly assigned in equal numbers to a placebo, a half dose and a full dose of a drug, with results

	improved	no difference	worse	
placebo	18	17	15	50
half dose	20	10	20	50
full dose	25	13	12	50
	63	40	47	150

Here the row totals are fixed at 50, so this is a test of homogeneity. The expected counts are $e_{ij} = 50 \times N_{+j}/150 = N_{+j}/3$, giving the first row of expectations (21, 13.33, 15.67) and similarly for the others. Computing,

$$2 \log \Lambda = 5.129, \quad \sum_{i,j} \frac{(o_{ij} - e_{ij})^2}{e_{ij}} = 5.173.$$

With $(r-1)(c-1) = 4$ degrees of freedom and $\chi_4^2(0.05) = 9.488$, we do not reject H_0 at the 5% level: this trial provides no significant evidence that the drug does anything.

Remark. Two weaknesses of these tests are worth flagging.

- (i) The χ^2 approximation needs n large; the usual heuristic is that every *expected* count should be at least 5. When it is not, one should use an exact test (such as Fisher's exact test), which is non-examinable.
- (ii) The tests often have low power, because Θ_1 is enormous. A test that must guard against every possible departure from H_0 cannot be very sensitive to any particular one.

§5.9 Tests and Confidence Sets

We close this section by making precise a duality that has been lurking throughout: confidence sets and hypothesis tests are two views of the same object.

Theorem 5.25

Let $X \sim f_X(\cdot | \theta)$, $\theta \in \Theta$.

- (i) Suppose that for each $\theta_0 \in \Theta$ there is a size- α test of $H_0 : \theta = \theta_0$ with acceptance region $A(\theta_0)$. Then

$$I(X) = \{\theta : X \in A(\theta)\}$$

is a $100(1 - \alpha)\%$ confidence set for θ .

- (ii) Conversely, if $I(X)$ is a $100(1 - \alpha)\%$ confidence set for θ , then $A(\theta_0) = \{x : \theta_0 \in I(x)\}$ is the acceptance region of a size- α test of $H_0 : \theta = \theta_0$.

Proof. Both parts follow from the single observation that, by construction,

$$\theta_0 \in I(X) \iff X \in A(\theta_0),$$

so the two events have the same probability under $\mathbb{P}_{\theta_0}$. For (i), the test having size α means $\mathbb{P}_{\theta_0}(X \in A(\theta_0)) = 1 - \alpha$, whence $\mathbb{P}_{\theta_0}(\theta_0 \in I(X)) = 1 - \alpha$; as θ_0 was arbitrary this is exactly the confidence set property. Part (ii) is the same computation read backwards. $\square$

In words – a confidence set is the collection of parameter values that the data would not have rejected. This is often the quickest route to a test, or to an interval.

Example 5.26

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known. We found the $100(1 - \alpha)\%$ confidence interval

$$I(X) = \left(\bar{X} \pm \frac{z_{\alpha/2} \sigma_0}{\sqrt{n}} \right).$$

By part (ii) of the theorem, a size- α test of $H_0 : \mu = \mu_0$ is obtained by rejecting when $\mu_0 \notin I(X)$, that is, when

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} \right| > z_{\alpha/2},$$

which is precisely the two-sided test we derived from the generalised likelihood ratio. The two derivations agree, as they must.

§6 Normal Theory Tests

Every test we have constructed so far has assumed the variance known, which is unrealistic. In this section we develop the distribution theory that lets us drop that assumption, and then deploy it in the standard tests for normal data. Everything rests on one theorem – the joint distribution of $\bar{X}$ and S_{XX} – which we prove in two ways.

In this section vectors are columns, and we write $u^\top$ for the transpose.

§6.1 The Multivariate Normal Distribution

We recall the essentials from IA Probability. For a random vector $X = (X_1, \dots, X_n)^\top$ we write $\mathbb{E}[X]$ for the vector of means and

$$\text{Var}(X) = \mathbb{E} \left[(X - \mathbb{E}X)(X - \mathbb{E}X)^\top \right]$$

for the covariance matrix, so $\text{Var}(X)_{ij} = \text{Cov}(X_i, X_j)$. For constant $A \in \mathbb{R}^{k \times n}$ and $b \in \mathbb{R}^k$ we have the familiar

$$\mathbb{E}[AX + b] = A \mathbb{E}[X] + b, \quad \text{Var}(AX + b) = A \text{Var}(X) A^\top.$$

Definition 6.1 (Multivariate Normal)

A random vector X has a **multivariate normal** distribution if $t^\top X$ is univariate normal for every fixed $t \in \mathbb{R}^n$ (allowing degenerate normals of zero variance).

Proposition 6.2

If X is multivariate normal then so is $AX + b$, for any constant $A \in \mathbb{R}^{k \times n}$ and $b \in \mathbb{R}^k$.

Proof. For $t \in \mathbb{R}^k$, $t^\top(AX + b) = (A^\top t)^\top X + t^\top b$ is a univariate normal plus a constant. $\square$

Proposition 6.3

A multivariate normal distribution is determined by its mean vector μ and covariance matrix Σ , and we write $X \sim N(\mu, \Sigma)$.

Proof. The moment generating function is $M_X(t) = \mathbb{E} \left[e^{t^\top X} \right]$. Since $t^\top X$ is univariate normal with mean $t^\top \mu$ and variance $t^\top \Sigma t$, evaluating the one-dimensional normal mgf at 1 gives

$$M_X(t) = \exp \left(t^\top \mu + \frac{1}{2} t^\top \Sigma t \right),$$

which depends only on μ and Σ . By uniqueness of moment generating functions, the distribution is determined. $\square$

An immediate corollary, which we will use constantly, is that for multivariate normal vectors *zero covariance implies independence*: if Σ is block diagonal, the mgf factorises accordingly.

§6.2 Orthogonal Projections

The geometry underlying normal theory is that of orthogonal projection, so we set up the linear algebra we need.

Definition 6.4 (Orthogonal Projection)

A matrix $P \in \mathbb{R}^{n \times n}$ is an **orthogonal projection** onto its column space $\text{col}(P)$ if $Pv = v$ for all $v \in \text{col}(P)$ and $Pw = 0$ for all $w \in \text{col}(P)^\perp$.

Proposition 6.5

P is an orthogonal projection if and only if P is symmetric and idempotent ($P^2 = P$).

Proof. Suppose P is symmetric and idempotent. If $v \in \text{col}(P)$ then $v = Pa$ for some a , so $Pv = P^2a = Pa = v$. If $w \in \text{col}(P)^\perp$ then $P^\top w = 0$ by definition of the orthogonal complement, and $P^\top = P$, so $Pw = 0$.

Conversely, suppose P is an orthogonal projection. Any $a \in \mathbb{R}^n$ decomposes uniquely as $a = v + w$ with $v \in \text{col}(P)$, $w \in \text{col}(P)^\perp$, and then $P^2a = P^2v + P^2w = Pv = P(v + w) = Pa$; as a was arbitrary, $P^2 = P$. For symmetry, note that for any u_1, u_2 we have $Pu_1 \in \text{col}(P)$ and $(I - P)u_2 \in \text{col}(P)^\perp$, so

$$u_1^\top P^\top (I - P)u_2 = (Pu_1) \cdot ((I - P)u_2) = 0.$$

Hence $P^\top (I - P) = 0$, i.e. $P^\top = P^\top P$. But $P^\top P$ is symmetric, so $P^\top$ is symmetric, so P is. $\square$

Corollary 6.6

If P is an orthogonal projection then so is $I - P$, projecting onto $\text{col}(P)^\perp$.

Proof. $I - P$ is symmetric, and $(I - P)^2 = I - 2P + P^2 = I - P$. $\square$

Proposition 6.7

If P is an orthogonal projection then $P = UU^\top$, where the columns of U form an orthonormal basis of $\text{col}(P)$. Consequently $\text{rank } P = \text{tr } P$.

Proof. Let U have as its columns an orthonormal basis of $\text{col}(P)$, so $U^\top U = I_r$ where $r = \dim \text{col}(P)$. Then $UU^\top$ is symmetric, and $(UU^\top)(UU^\top) = U(U^\top U)U^\top = UU^\top$, so it is an orthogonal projection; its column space is $\text{col}(U) = \text{col}(P)$, and two orthogonal projections with the same column space are equal (both act as the identity there and as zero on the complement).

For the rank, $\text{rank } P = r = \text{tr}(I_r) = \text{tr}(U^\top U) = \text{tr}(UU^\top) = \text{tr } P$, using $\text{tr}(AB) = \text{tr}(BA)$. $\square$

Now the key distributional result.

Theorem 6.8 (Projections of Spherical Normals)

Let $X \sim N(0, \sigma^2 I_n)$ and let P be an orthogonal projection of rank r . Then

- (i) $PX \sim N(0, \sigma^2 P)$ and $(I - P)X \sim N(0, \sigma^2(I - P))$, and these are independent;

$$(ii) \|PX\|^2/\sigma^2 \sim \chi_r^2.$$

Proof. The stacked vector $(PX, (I - P)X)^\top \in \mathbb{R}^{2n}$ is a linear function of X , hence multivariate normal, and is therefore determined by its mean and covariance. Its mean is 0, and its covariance is

$$\begin{pmatrix} P \\ I - P \end{pmatrix} \sigma^2 I \begin{pmatrix} P \\ I - P \end{pmatrix}^\top = \sigma^2 \begin{pmatrix} P^2 & P(I - P) \\ (I - P)P & (I - P)^2 \end{pmatrix} = \sigma^2 \begin{pmatrix} P & 0 \\ 0 & I - P \end{pmatrix},$$

using idempotence and $P(I - P) = P - P^2 = 0$. This identifies the marginals, and since the covariance is block diagonal and the vector is multivariate normal, the two blocks are independent.

For (ii), write $P = UU^\top$ as above. Then

$$\|PX\|^2 = X^\top P^\top PX = X^\top PX = X^\top UU^\top X = \|U^\top X\|^2,$$

and $U^\top X \sim N(0, \sigma^2 U^\top U) = N(0, \sigma^2 I_r)$. So the r components of $U^\top X/\sigma$ are i.i.d $N(0, 1)$, and

$$\frac{\|PX\|^2}{\sigma^2} = \sum_{i=1}^r \left(\frac{(U^\top X)_i}{\sigma} \right)^2 \sim \chi_r^2. \quad \square$$

§6.3 Normal Random Samples

We can now prove the central result of the section.

Theorem 6.9 (Joint Distribution of $\bar{X}$ and S_{XX})

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$. Then

- (i) $\bar{X} \sim N(\mu, \sigma^2/n)$;
- (ii) $S_{XX}/\sigma^2 \sim \chi_{n-1}^2$;
- (iii) $\bar{X}$ and S_{XX} are independent.

Proof. Let P be the $n \times n$ matrix with every entry $1/n$. It is symmetric, and $P^2 = P$ since each entry of P^2 is $n \cdot \frac{1}{n^2} = \frac{1}{n}$; so P is an orthogonal projection, onto the one-dimensional space of constant vectors. Note $PX = (\bar{X}, \dots, \bar{X})^\top$ and $\text{tr } P = 1$, so $\text{rank } P = 1$ and $\text{rank}(I - P) = n - 1$.

Write $X = M + \varepsilon$ where $M = (\mu, \dots, \mu)^\top$ and $\varepsilon \sim N(0, \sigma^2 I)$. Since M is constant, $PM = M$ and $(I - P)M = 0$, so

$$\bar{X} = (PX)_1 = \mu + (P\varepsilon)_1$$

is a function of $P\varepsilon$, while

$$S_{XX} = \sum_i (X_i - \bar{X})^2 = \|X - PX\|^2 = \|(I - P)X\|^2 = \|(I - P)\varepsilon\|^2$$

is a function of $(I - P)\varepsilon$. By Theorem ?? these two are independent, giving (iii),

and

$$\frac{S_{XX}}{\sigma^2} = \frac{\|(I - P)\varepsilon\|^2}{\sigma^2} \sim \chi_{n-1}^2,$$

giving (ii). Finally $\bar{X}$ is normal (a linear function of a normal vector) with mean μ and variance σ^2/n , giving (i). $\square$

Remark. This is where the mysterious $n - 1$ of Section 1 comes from. The data live in $\mathbb{R}^n$; fitting the mean uses up the one-dimensional space of constant vectors, and the residual $X - \bar{X}\mathbf{1}$ is confined to the orthogonal complement, of dimension $n - 1$. The ‘degrees of freedom’ really are dimensions of a subspace. It also explains why $\bar{X}$ and S_{XX} are independent, which is otherwise a rather surprising fact: they are functions of the components of ε in orthogonal subspaces.

There is a classical alternative proof, which is worth seeing because it makes the change of coordinates explicit.

Alternative proof, by an orthogonal change of basis. Let A be any $n \times n$ orthogonal matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$; such a matrix exists by extending that unit vector to an orthonormal basis. Put $Y = AX$, so

$$Y \sim N(AM, \sigma^2 AA^\top) = N(AM, \sigma^2 I),$$

and $AM = (\sqrt{n}\mu, 0, \dots, 0)^\top$. So $Y_1 \sim N(\sqrt{n}\mu, \sigma^2)$ and $Y_2, \dots, Y_n$ are i.i.d $N(0, \sigma^2)$, all independent. But $Y_1 = \frac{1}{\sqrt{n}} \sum_i X_i = \sqrt{n}\bar{X}$, giving (i), and

$$Y_2^2 + \dots + Y_n^2 = \|Y\|^2 - Y_1^2 = X^\top A^\top A X - n\bar{X}^2 = \sum_i X_i^2 - n\bar{X}^2 = S_{XX},$$

a sum of $n - 1$ independent $N(0, \sigma^2)$ squares, giving (ii). Independence of Y_1 from $Y_2, \dots, Y_n$ gives (iii). $\square$

§6.4 The One-Sample t -Test

Now we can finally handle unknown variance. Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$ with both parameters unknown, and write

$$\tilde{\sigma}^2 = \frac{S_{XX}}{n-1}$$

for the unbiased estimator of σ^2 . By Theorem ??,

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1), \quad V = \frac{S_{XX}}{\sigma^2} \sim \chi_{n-1}^2,$$

and these are independent. So by the definition of the t -distribution,

$$\frac{Z}{\sqrt{V/(n-1)}} = \frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{\sqrt{S_{XX}/((n-1)\sigma^2)}} = \frac{\sqrt{n}(\bar{X} - \mu)}{\tilde{\sigma}} \sim t_{n-1}.$$

The unknown σ has cancelled: this is a genuine pivot, and it is the reason the t -distribution was invented.

Proposition 6.10 (Confidence Interval for a Normal Mean)

With $X_1, \dots, X_n$ i.i.d $N(\mu, \sigma^2)$, both unknown, a $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm \frac{\tilde{\sigma}}{\sqrt{n}} t_{n-1} \left(\frac{\alpha}{2} \right).$$

Proof. Immediate from the pivot above and the symmetry of t_{n-1} about 0:

$$1 - \alpha = \mathbb{P} \left(-t_{n-1} \left(\frac{\alpha}{2} \right) \leq \frac{\sqrt{n}(\bar{X} - \mu)}{\tilde{\sigma}} \leq t_{n-1} \left(\frac{\alpha}{2} \right) \right),$$

and rearranging for μ . □

Definition 6.11 (One-Sample t -Test)

To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ with σ^2 unknown, compute

$$t = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\tilde{\sigma}}$$

and reject H_0 at size α if $|t| > t_{n-1}(\alpha/2)$. For the one-sided alternative $H_1 : \mu > \mu_0$ we reject if $t > t_{n-1}(\alpha)$.

By the duality of Section 5.9 this is exactly the test that rejects when μ_0 falls outside the confidence interval above. Note that $t_{n-1}(\alpha/2) > z_{\alpha/2}$, so the interval is wider and the test more conservative than the corresponding Z -procedures: this is the price of not knowing σ .

Example 6.12

Suppose $n = 10$ observations give $\bar{x} = 4.2$ and $\tilde{\sigma} = 1.1$, and we test $H_0 : \mu = 3.5$ against $H_1 : \mu \neq 3.5$. Then

$$t = \frac{\sqrt{10}(4.2 - 3.5)}{1.1} = \frac{3.162 \times 0.7}{1.1} \approx 2.01.$$

Since $t_9(0.025) = 2.262$, we do not reject at the 5% level – though we would at the 10% level, as $t_9(0.05) = 1.833$.

§6.5 The Two-Sample t -Test

More often we wish to compare two groups. Let

$$X_1, \dots, X_m \stackrel{\text{iid}}{\sim} N(\mu_X, \sigma^2), \quad Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu_Y, \sigma^2)$$

be independent samples with a *common* unknown variance, and test $H_0 : \mu_X = \mu_Y$.

The natural estimator of the difference is $\bar{X} - \bar{Y}$, with

$$\bar{X} - \bar{Y} \sim N \left(\mu_X - \mu_Y, \sigma^2 \left(\frac{1}{m} + \frac{1}{n} \right) \right).$$

For the variance we pool the two sums of squares. Since $S_{XX}/\sigma^2 \sim \chi_{m-1}^2$ and $S_{YY}/\sigma^2 \sim \chi_{n-1}^2$ independently, and independent chi-squareds add,

$$\frac{S_{XX} + S_{YY}}{\sigma^2} \sim \chi_{m+n-2}^2,$$

and this is independent of $\bar{X} - \bar{Y}$ (each sum of squares is independent of its own sample mean, and the samples are independent of each other). Writing

$$\tilde{\sigma}^2 = \frac{S_{XX} + S_{YY}}{m + n - 2}$$

for the **pooled** variance estimator, we obtain the pivot

$$\frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\tilde{\sigma} \sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}.$$

Definition 6.13 (Two-Sample *t*-Test)

Under the above assumptions, to test $H_0 : \mu_X = \mu_Y$ compute

$$t = \frac{\bar{X} - \bar{Y}}{\tilde{\sigma} \sqrt{\frac{1}{m} + \frac{1}{n}}}$$

and reject at size α if $|t| > t_{m+n-2}(\alpha/2)$. Correspondingly,

$$(\bar{X} - \bar{Y}) \pm \tilde{\sigma} \sqrt{\frac{1}{m} + \frac{1}{n}} t_{m+n-2}\left(\frac{\alpha}{2}\right)$$

is a $100(1 - \alpha)\%$ confidence interval for $\mu_X - \mu_Y$.

The assumption of equal variances is essential to the pooling step. If it is doubtful, one should first test it (see below), or use an approximate procedure such as Welch's test, which is not part of this course.

§6.6 Paired Data

Suppose instead that the observations come in pairs (X_i, Y_i) , $i = 1, \dots, n$ – the same patient measured before and after treatment, say, or two measuring instruments applied to the same specimen. The two-sample test is then *wrong*, because it assumes the samples independent, and here they are emphatically not: a patient with a high ‘before’ reading will tend to have a high ‘after’ reading too.

The remedy is disarmingly simple. Set $D_i = X_i - Y_i$ and treat $D_1, \dots, D_n$ as a single sample. If the pairs are i.i.d then so are the D_i , and testing $H_0 : \mathbb{E}[D] = 0$ by the one-sample *t*-test on the differences is exactly right.

Definition 6.14 (Paired *t*-Test)

Given paired data, let $D_i = X_i - Y_i$ and suppose $D_1, \dots, D_n$ are i.i.d $N(\mu_D, \sigma_D^2)$.

To test $H_0 : \mu_D = 0$, compute

$$t = \frac{\sqrt{n} \bar{D}}{\tilde{\sigma}_D}, \quad \tilde{\sigma}_D^2 = \frac{1}{n-1} \sum_i (D_i - \bar{D})^2,$$

and reject at size α if $|t| > t_{n-1}(\alpha/2)$.

Pairing is not merely a technical correction; it is usually a large *gain*. The subject-to-subject variability, which in the two-sample test is noise inflating $\tilde{\sigma}$, cancels entirely in the differences. When a paired design is available it will almost always detect a given effect with far fewer observations than an unpaired one.

Example 6.15

Eight subjects have their reaction times measured before and after a dose of caffeine, with differences (before minus after, in milliseconds)

$$D = (12, -3, 18, 7, 22, 5, 14, 9).$$

Here $\bar{D} = 10.5$ and $\sum_i (D_i - \bar{D})^2 = 430$, so $\tilde{\sigma}_D^2 = 430/7 \approx 61.4$ and $\tilde{\sigma}_D \approx 7.84$. Then

$$t = \frac{\sqrt{8} \times 10.5}{7.84} \approx 3.79,$$

against $t_7(0.025) = 2.365$. So we reject H_0 at the 5% level and conclude that caffeine has an effect.

§6.7 The F -Test for Equality of Variances

Finally, suppose we wish to compare two variances – either because it is the question of interest, or to check the pooling assumption of the two-sample t -test. Let $X_1, \dots, X_m$ be i.i.d $N(\mu_X, \sigma_X^2)$ and $Y_1, \dots, Y_n$ be i.i.d $N(\mu_Y, \sigma_Y^2)$, independent, and test $H_0 : \sigma_X^2 = \sigma_Y^2$.

We know $S_{XX}/\sigma_X^2 \sim \chi_{m-1}^2$ and $S_{YY}/\sigma_Y^2 \sim \chi_{n-1}^2$, independently. So by definition of the F -distribution,

$$\frac{S_{XX}/((m-1)\sigma_X^2)}{S_{YY}/((n-1)\sigma_Y^2)} = \frac{\tilde{\sigma}_X^2/\sigma_X^2}{\tilde{\sigma}_Y^2/\sigma_Y^2} \sim F_{m-1, n-1}.$$

This is a pivot for the ratio σ_X^2/σ_Y^2 , so it gives both a test and a confidence interval.

Definition 6.16 (F -Test for Equality of Variances)

Under $H_0 : \sigma_X^2 = \sigma_Y^2$ the statistic

$$F = \frac{\tilde{\sigma}_X^2}{\tilde{\sigma}_Y^2}$$

has the $F_{m-1, n-1}$ distribution. We reject H_0 at size α against the two-sided alternative if $F > F_{m-1, n-1}(\alpha/2)$ or $F < F_{m-1, n-1}(1 - \alpha/2)$.

Remark. Recall that $1/F \sim F_{n-1, m-1}$, so one may always arrange to have the larger sample variance on top and use only the upper tail (doubling the resulting p -value for a two-sided test). Note also that this test is notoriously sensitive to departures from

normality – far more so than the t -test, which is quite robust. A significant F here may be telling us about the tails of the data rather than about the variances.

Example 6.17

Suppose $m = 11$ observations give $\tilde{\sigma}_X^2 = 24.5$, and $n = 16$ give $\tilde{\sigma}_Y^2 = 9.8$. Then $F = 24.5/9.8 = 2.5$ on $(10, 15)$ degrees of freedom. Since $F_{10,15}(0.025) = 3.06$, we do not reject equality of variances at the 5% level, and pooling for a subsequent t -test would be defensible.

§7 The Linear Model

The linear model is the single most used statistical model in existence, and it is a fitting place to end, because it brings together everything we have done: least squares as maximum likelihood, projections and degrees of freedom, t -tests for individual parameters, F -tests from the generalised likelihood ratio, and confidence intervals from pivots.

§7.1 The Model

Suppose we observe pairs $(x_1, Y_1), \dots, (x_n, Y_n)$ where $Y_i \in \mathbb{R}$ and $x_i = (x_{i1}, \dots, x_{ip})^\top \in \mathbb{R}^p$. The Y_i are the **responses** (or dependent variables); the x_{ij} are the **predictors** (or covariates, or independent variables) and are treated as fixed and known. We model

$$Y_i = \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i,$$

where $\beta \in \mathbb{R}^p$ are unknown **coefficients** and ε is a random **noise** vector.

Example 7.1

Let Y_i be the number of insurance claims made by driver i in a year, and let the predictors record the driver's age, years of experience, and points on their licence. The model asserts that the expected number of claims is a linear combination of these.

An **intercept** is included by setting $x_{i1} = 1$ for all i , so that β_1 is a constant term; we will always assume this has been done unless stated otherwise. It is worth stressing that 'linear' refers to linearity in β , not in the predictors: the model

$$Y_i = a + bz_i + cz_i^2 + \varepsilon_i$$

is a perfectly good linear model with $x_i = (1, z_i, z_i^2)^\top$. This flexibility is a large part of why the model is so widely used.

The coefficient β_j is interpreted as the change in $\mathbb{E}[Y_i]$ produced by increasing x_{ij} by one while holding all other predictors fixed. This is emphatically *not* a causal statement unless the predictors were assigned by the experimenter – a point on which a great deal of applied statistics comes to grief.

§7.2 Matrix Formulation and Assumptions

Stacking everything up, let

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad X = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

where X is called the **design matrix**. The model is then simply

$$Y = X\beta + \varepsilon.$$

We make two **moment assumptions** on the noise:

- (i) $\mathbb{E}[\varepsilon] = 0$, so that $\mathbb{E}[Y_i] = x_i^\top \beta$ – the model is correctly specified;
- (ii) $\text{Var}(\varepsilon) = \sigma^2 I$, that is, $\text{Var}(\varepsilon_i) = \sigma^2$ for all i and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. Constant variance is called **homoscedasticity**.

Note that no distributional assumption has yet been made; we will add normality in Section 7.5.

Throughout we assume the design matrix X has full rank p , equivalently that its columns are linearly independent. Since $X \in \mathbb{R}^{n \times p}$ this forces $n \geq p$: we need at least as many observations as parameters, which is only sensible.

§7.3 Least Squares Estimation

Definition 7.2 (Least Squares Estimator)

The **least squares estimator** $\hat{\beta}$ minimises the **residual sum of squares**

$$S(\beta) = \|Y - X\beta\|^2 = \sum_{i=1}^n \left(Y_i - x_i^\top \beta \right)^2.$$

Proposition 7.3

If X has full column rank then S has the unique minimiser

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Moreover $\mathbb{E}[\hat{\beta}] = \beta$ and $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$.

Proof. S is a positive semi-definite quadratic in β , so is minimised at a stationary point. Differentiating,

$$\frac{\partial S}{\partial \beta_k} = -2 \sum_{i=1}^n x_{ik} \left(Y_i - \sum_j x_{ij} \beta_j \right),$$

and setting all p of these to zero gives the **normal equations**

$$X^\top X \hat{\beta} = X^\top Y.$$

Since X has full column rank, $X^\top X$ is invertible: if $X^\top Xv = 0$ then $\|Xv\|^2 = v^\top X^\top Xv = 0$, so $Xv = 0$, so $v = 0$. Hence $\hat{\beta} = (X^\top X)^{-1}X^\top Y$, and this is the unique stationary point.

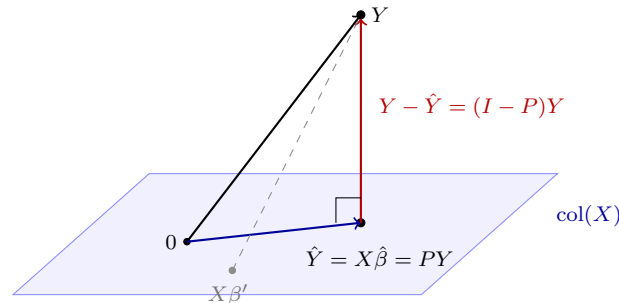
As $\hat{\beta}$ is a linear function of Y ,

$$\mathbb{E}[\hat{\beta}] = (X^\top X)^{-1}X^\top \mathbb{E}[Y] = (X^\top X)^{-1}X^\top X\beta = \beta,$$

and writing $B = (X^\top X)^{-1}X^\top$,

$$\text{Var}(\hat{\beta}) = B \text{Var}(Y)B^\top = \sigma^2 BB^\top = \sigma^2 (X^\top X)^{-1}X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}. \quad \square$$

The geometry is much more illuminating than the algebra. The vector $X\beta$ ranges over $\text{col}(X)$, a p -dimensional subspace of $\mathbb{R}^n$, as β ranges over $\mathbb{R}^p$. Minimising $\|Y - X\beta\|$ therefore means finding the point of $\text{col}(X)$ closest to Y – which is exactly the orthogonal projection of Y onto $\text{col}(X)$.



Any other point $X\beta'$ of the subspace is further from Y , by Pythagoras. Let us record this projection formally.

Definition 7.4 (Fitted Values, Hat Matrix, Residuals)

The **fitted values** are $\hat{Y} = X\hat{\beta} = PY$ where

$$P = X(X^\top X)^{-1}X^\top$$

is the **hat matrix**. The **residuals** are $Y - \hat{Y} = (I - P)Y$.

Proposition 7.5

P is the orthogonal projection onto $\text{col}(X)$.

Proof. If $v \in \text{col}(X)$ then $v = Xb$ for some b , and $Pv = X(X^\top X)^{-1}X^\top Xb = Xb = v$. If $w \in \text{col}(X)^\perp$ then $X^\top w = 0$, so $Pw = 0$. (Equivalently: P is visibly symmetric, and $P^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = P$.) $\square$

In particular $\text{tr } P = \text{rank } P = p$ and $\text{rank}(I - P) = n - p$, and the residuals are orthogonal to every column of X . That last fact is worth remembering: least squares residuals are, by construction, uncorrelated with every predictor in the model.

§7.4 The Gauss-Markov Theorem

Least squares is not just geometrically natural; among a large class of estimators it is optimal, and remarkably this requires no normality assumption at all.

Theorem 7.6 (Gauss-Markov)

Let $\beta^* = CY$ be any estimator of β that is linear in Y and unbiased. Then for every $t \in \mathbb{R}^p$,

$$\text{Var}(t^\top \hat{\beta}) \leq \text{Var}(t^\top \beta^*),$$

where $\hat{\beta}$ is the least squares estimator. We say $\hat{\beta}$ is the **best linear unbiased estimator** (BLUE).

Remark. Thinking of t as the predictor vector for a new observation, $t^\top \hat{\beta}$ is our prediction of its expected response. The theorem says least squares gives the most precise such prediction among all linear unbiased methods, simultaneously for every t . Taking $t = e_j$ recovers the statement that $\text{Var}(\hat{\beta}_j) \leq \text{Var}(\beta_j^*)$ for each coordinate.

Proof. Since $\text{Var}(t^\top \beta^*) - \text{Var}(t^\top \hat{\beta}) = t^\top [\text{Var}(\beta^*) - \text{Var}(\hat{\beta})] t$, it suffices to show that $\text{Var}(\beta^*) - \text{Var}(\hat{\beta})$ is positive semi-definite.

Put $A = C - (X^\top X)^{-1} X^\top$, so that $\beta^* = \hat{\beta} + AY$. Unbiasedness of both estimators gives $\mathbb{E}[AY] = 0$; but $\mathbb{E}[AY] = AX\beta$, and as this vanishes for every β we must have $AX = 0$. Now

$$\begin{aligned} \text{Var}(\beta^*) &= \text{Var}\left(\left[A + (X^\top X)^{-1} X^\top\right] Y\right) \\ &= \sigma^2 \left[A + (X^\top X)^{-1} X^\top\right] \left[A + (X^\top X)^{-1} X^\top\right]^\top \\ &= \sigma^2 \left(AA^\top + \underbrace{AX(X^\top X)^{-1}}_{=0} + (X^\top X)^{-1} \underbrace{X^\top A^\top}_{=0} + (X^\top X)^{-1}\right) \\ &= \sigma^2 AA^\top + \text{Var}(\hat{\beta}). \end{aligned}$$

So $\text{Var}(\beta^*) - \text{Var}(\hat{\beta}) = \sigma^2 AA^\top$, and $u^\top AA^\top u = \|A^\top u\|^2 \geq 0$ for every u . $\square$

§7.5 The Normal Linear Model

We now strengthen the moment assumptions to a full distributional one.

Definition 7.7 (Normal Linear Model)

The **normal linear model** is $Y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$, where β and $\sigma^2 > 0$ are both unknown.

The likelihood is

$$L(\beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \|Y - X\beta\|^2\right),$$

so

$$\ell(\beta, \sigma^2) = \text{const} - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|Y - X\beta\|^2.$$

For any fixed σ^2 , maximising over β means minimising $\|Y - X\beta\|^2$. So:

Proposition 7.8

In the normal linear model the maximum likelihood estimator of β is the least squares estimator $\hat{\beta} = (X^\top X)^{-1} X^\top Y$, and

$$\hat{\sigma}^2 = \frac{1}{n} \|Y - X\hat{\beta}\|^2 = \frac{1}{n} \|(I - P)Y\|^2.$$

Proof. The first claim is the observation above. Substituting $\hat{\beta}$ and differentiating in σ^2 ,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \|(I - P)Y\|^2,$$

which vanishes at the stated value, and this is a maximum. $\square$

That least squares and maximum likelihood coincide under normal errors is a happy accident of the exponential of a quadratic, but it is the reason least squares is so ubiquitous.

Theorem 7.9 (Distribution Theory for the Normal Linear Model)

In the normal linear model,

- (i) $\hat{\beta} \sim N(\beta, \sigma^2(X^\top X)^{-1})$;
- (ii) $n\hat{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$;
- (iii) $\hat{\beta}$ and $\hat{\sigma}^2$ are independent.

Proof. (i) We have already computed the mean and variance of $\hat{\beta}$, and it is a linear function of the multivariate normal Y , hence multivariate normal.

(ii) Since P projects onto $\text{col}(X)$ we have $(I - P)X = 0$, so

$$\frac{n\hat{\sigma}^2}{\sigma^2} = \frac{\|(I - P)Y\|^2}{\sigma^2} = \frac{\|(I - P)(X\beta + \varepsilon)\|^2}{\sigma^2} = \frac{\|(I - P)\varepsilon\|^2}{\sigma^2} \sim \chi_{\text{rank}(I - P)}^2$$

by Theorem ??, and $\text{rank}(I - P) = \text{tr } I - \text{tr } P = n - p$.

(iii) From (ii), $\hat{\sigma}^2$ is a function of $(I - P)\varepsilon$. On the other hand

$$\hat{\beta} = (X^\top X)^{-1} X^\top (X\beta + \varepsilon) = \beta + (X^\top X)^{-1} X^\top \varepsilon = \beta + (X^\top X)^{-1} X^\top P\varepsilon,$$

using $X^\top P = X^\top$, so $\hat{\beta}$ is a function of $P\varepsilon$. These are independent by Theorem ?? $\square$

Since $\mathbb{E}[\chi_{n-p}^2] = n - p$ we get

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n - p}{n} \sigma^2 < \sigma^2,$$

so as usual the maximum likelihood estimator of the variance is biased downwards. We therefore define the unbiased estimator

$$\tilde{\sigma}^2 = \frac{n\hat{\sigma}^2}{n - p} = \frac{\|(I - P)Y\|^2}{n - p},$$

and note that $\tilde{\sigma}^2$ is what one should always use in practice. Once again the divisor is the dimension of the space in which the residuals live.

§7.6 Inference for the Coefficients

Theorem ?? gives us everything we need to construct pivots.

Proposition 7.10 (Confidence Interval for a Single Coefficient)

In the normal linear model,

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\tilde{\sigma}^2 (X^\top X)_{jj}^{-1}}} \sim t_{n-p},$$

and hence

$$\hat{\beta}_j \pm t_{n-p} \left(\frac{\alpha}{2} \right) \sqrt{\tilde{\sigma}^2 (X^\top X)_{jj}^{-1}}$$

is a $100(1 - \alpha)\%$ confidence interval for β_j .

Proof. By part (i) of Theorem ??, the j th coordinate is distributed as

$$\hat{\beta}_j \sim N \left(\beta_j, \sigma^2 (X^\top X)_{jj}^{-1} \right),$$

so

$$U = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{(X^\top X)_{jj}^{-1}}} \sim N(0, 1).$$

By parts (ii) and (iii), $V = n\hat{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$ independently of U . Hence

$$\frac{U}{\sqrt{V/(n-p)}} = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{(X^\top X)_{jj}^{-1}}} \cdot \frac{\sigma}{\tilde{\sigma}} = \frac{\hat{\beta}_j - \beta_j}{\tilde{\sigma} \sqrt{(X^\top X)_{jj}^{-1}}} \sim t_{n-p},$$

the unknown σ having cancelled. Rearranging the two-sided t inequality gives the interval. $\square$

By the test-interval duality, a size- α test of $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$ rejects exactly when 0 falls outside this interval, i.e. when

$$\left| \frac{\hat{\beta}_j}{\tilde{\sigma} \sqrt{(X^\top X)_{jj}^{-1}}} \right| > t_{n-p} \left(\frac{\alpha}{2} \right).$$

This is the t -statistic that statistical software prints next to every coefficient.

Example 7.11 (The One-Sample t -Test as a Linear Model)

Take $p = 1$ with $X = (1, \dots, 1)^\top$, so the model is $Y_i = \mu + \varepsilon_i$, i.e. $Y_1, \dots, Y_n$ i.i.d $N(\mu, \sigma^2)$. Then $X^\top X = n$, $\hat{\beta} = \bar{Y}$, $(X^\top X)_{11}^{-1} = 1/n$ and $\tilde{\sigma}^2 = S_{YY}/(n-1)$, so the interval above is exactly

$$\bar{Y} \pm \frac{\tilde{\sigma}}{\sqrt{n}} t_{n-1} \left(\frac{\alpha}{2} \right),$$

recovering the one-sample t -interval of Section 6.4.

We can also give a confidence set for the whole vector β at once.

Proposition 7.12 (Confidence Set for β)

In the normal linear model,

$$\frac{\|X(\hat{\beta} - \beta)\|^2 / p}{\tilde{\sigma}^2} \sim F_{p, n-p},$$

so

$$\left\{ \beta' \in \mathbb{R}^p : \frac{\|X(\hat{\beta} - \beta')\|^2}{p \tilde{\sigma}^2} \leq F_{p, n-p}(\alpha) \right\}$$

is a $100(1 - \alpha)\%$ confidence set for β .

Proof. Since $\hat{\beta} - \beta \sim N(0, \sigma^2(X^\top X)^{-1})$, writing $(X^\top X)^{1/2}$ for the symmetric positive definite square root,

$$(X^\top X)^{1/2}(\hat{\beta} - \beta) \sim N\left(0, \sigma^2(X^\top X)^{1/2}(X^\top X)^{-1}(X^\top X)^{1/2}\right) = N(0, \sigma^2 I_p),$$

so its squared norm over σ^2 is χ_p^2 . But that squared norm is

$$(\hat{\beta} - \beta)^\top X^\top X (\hat{\beta} - \beta) = \|X(\hat{\beta} - \beta)\|^2.$$

This is a function of $\hat{\beta}$, hence independent of $\tilde{\sigma}^2$, and $(n - p)\tilde{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$. Taking the ratio of the two scaled chi-squareds gives the F distribution, the σ^2 cancelling. $\square$

Geometrically the confidence set is an ellipsoid centred at $\hat{\beta}$, with principal axes along the eigenvectors of $X^\top X$: directions in which the predictors carry little information give long axes. Note that both results in this subsection are *exact* – no asymptotics have been used anywhere.

§7.7 F -Tests and Model Comparison

Now suppose we wish to test whether a whole block of coefficients vanishes. Relabelling if necessary, take

$$H_0 : \beta_1 = \cdots = \beta_{p_0} = 0, \quad H_1 : \beta \in \mathbb{R}^p.$$

Split $X = (X_0, X_1)$ with $X_0 \in \mathbb{R}^{n \times p_0}$ and $X_1 \in \mathbb{R}^{n \times (p - p_0)}$, and correspondingly $\beta = (\beta^0, \beta^1)$. Under H_0 the model reduces to $Y = X_1 \beta^1 + \varepsilon$. Write

$$P = X(X^\top X)^{-1}X^\top, \quad P_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$$

for the projections onto $\text{col}(X)$ and the smaller $\text{col}(X_1)$.

Lemma 7.13

$P - P_1$ is an orthogonal projection of rank p_0 , and $(I - P)(P - P_1) = 0$.

Proof. Since $\text{col}(X_1) \subseteq \text{col}(X)$, applying P to anything in $\text{col}(X_1)$ leaves it fixed, so $PP_1 = P_1$; taking transposes, $P_1P = P_1$. Hence $P - P_1$ is symmetric, and

$$(P - P_1)^2 = P^2 - PP_1 - P_1P + P_1^2 = P - P_1 - P_1 + P_1 = P - P_1,$$

so it is an orthogonal projection. Its rank is $\text{tr } P - \text{tr } P_1 = p - (p - p_0) = p_0$. Finally

$$(I - P)(P - P_1) = P - P_1 - P^2 + PP_1 = P - P_1 - P + P_1 = 0. \quad \square$$

Now we compute the generalised likelihood ratio. From the proposition above, the maximised log-likelihood in a normal linear model with projection P is

$$\ell(\hat{\beta}, \hat{\sigma}^2) = \text{const} - \frac{n}{2} \log \frac{\|(I - P)Y\|^2}{n},$$

so

$$2 \log \Lambda = n \left[\log \frac{\|(I - P_1)Y\|^2}{n} - \log \frac{\|(I - P)Y\|^2}{n} \right] = n \log \frac{\|(I - P_1)Y\|^2}{\|(I - P)Y\|^2}.$$

This is increasing in the ratio of sums of squares, which by Lemma ?? and Pythagoras satisfies

$$\begin{aligned} \frac{\|(I - P_1)Y\|^2}{\|(I - P)Y\|^2} &= \frac{\|(I - P)Y + (P - P_1)Y\|^2}{\|(I - P)Y\|^2} \\ &= \frac{\|(I - P)Y\|^2 + \|(P - P_1)Y\|^2}{\|(I - P)Y\|^2} = 1 + \frac{\|(P - P_1)Y\|^2}{\|(I - P)Y\|^2}, \end{aligned}$$

the cross term vanishing because $(I - P)(P - P_1) = 0$. So the generalised likelihood ratio test rejects for large values of

$$F = \frac{\|(P - P_1)Y\|^2/p_0}{\|(I - P)Y\|^2/(n - p)},$$

and, delightfully, we can find its exact distribution rather than appealing to Wilks.

Theorem 7.14

Under $H_0 : \beta_1 = \dots = \beta_{p_0} = 0$ in the normal linear model,

$$F = \frac{\|(P - P_1)Y\|^2/p_0}{\|(I - P)Y\|^2/(n - p)} \sim F_{p_0, n-p}.$$

Proof. We know $\|(I - P)Y\|^2 = \|(I - P)\varepsilon\|^2 \sim \sigma^2 \chi_{n-p}^2$. Under H_0 we have $Y = X_1\beta^1 + \varepsilon$, and since both P and P_1 fix $\text{col}(X_1)$,

$$(P - P_1)Y = (P - P_1)X_1\beta^1 + (P - P_1)\varepsilon = (P - P_1)\varepsilon,$$

so by Theorem ?? and Lemma ??, $\|(P - P_1)Y\|^2 \sim \sigma^2 \chi_{p_0}^2$.

For independence, note that the stacked vector $((I - P)\varepsilon, (P - P_1)\varepsilon)$ is multivariate normal with mean zero and covariance

$$\sigma^2 \begin{pmatrix} (I - P)^2 & (I - P)(P - P_1) \\ (P - P_1)(I - P) & (P - P_1)^2 \end{pmatrix} = \sigma^2 \begin{pmatrix} I - P & 0 \\ 0 & P - P_1 \end{pmatrix},$$

which is block diagonal, so the two blocks are independent. Taking the ratio of the two scaled chi-squareds and cancelling σ^2 gives $F_{p_0, n-p}$. $\square$

So the size- α test rejects H_0 when $F > F_{p_0, n-p}(\alpha)$, and this is exact for every n, p, p_0 .

Remark. The numerator $\|(P - P_1)Y\|^2$ is the reduction in residual sum of squares achieved by adding the p_0 extra predictors, and the denominator estimates σ^2 . So F compares ‘variance explained per extra parameter’ against ‘unexplained variance per residual degree of freedom’. Adding predictors always reduces the residual sum of squares; the F -test asks whether it reduces it by more than chance would predict.

Proposition 7.15

When $p_0 = 1$, the F -test of $H_0 : \beta_1 = 0$ is identical to the two-sided t -test of the same hypothesis.

Proof sketch. Recall from Section 1 that $T \sim t_{n-p}$ implies $T^2 \sim F_{1, n-p}$, and correspondingly $t_{n-p}(\alpha/2)^2 = F_{1, n-p}(\alpha)$. So the t -test rejects exactly when

$$\frac{\hat{\beta}_1^2}{\tilde{\sigma}^2(X^\top X)^{-1}_{11}} > F_{1, n-p}(\alpha),$$

and since the denominators of the two statistics already agree, it suffices to check that

$$\frac{\hat{\beta}_1^2}{(X^\top X)^{-1}_{11}} = \|(P - P_1)Y\|^2.$$

When $p_0 = 1$, $P - P_1$ is the rank-one projection onto the span of $v = (I - P_1)X_0$, where X_0 is the first column of X . A computation using $X_0^\top(I - P_1) = v^\top = v^\top(P - P_1)$ gives

$$\|(P - P_1)Y\|^2 = \frac{(v^\top Y)^2}{\|v\|^2} = \frac{\left(X_0^\top(I - P_1)X\hat{\beta}\right)^2}{\|(I - P_1)X_0\|^2} = \|(I - P_1)X_0\|^2 \hat{\beta}_1^2,$$

using $(I - P_1)X = [(I - P_1)X_0, 0, \dots, 0]$. Finally, blockwise matrix inversion gives $(X^\top X)^{-1}_{11} = \|(I - P_1)X_0\|^{-2}$, and the two expressions agree. $\square$

§7.8 Simple Linear Regression

The simplest interesting case is $p = 2$ with a single predictor. It is customary to centre the predictor, writing

$$Y_i = a + b(x_i - \bar{x}) + \varepsilon_i, \quad \bar{x} = \frac{1}{n} \sum_i x_i,$$

which makes the algebra much cleaner because it renders the two columns of the design matrix orthogonal.

Proposition 7.16

In the centred simple linear regression model,

$$\hat{a} = \bar{Y}, \quad \hat{b} = \frac{S_{xY}}{S_{xx}},$$

where $S_{xY} = \sum_i (x_i - \bar{x})(Y_i - \bar{Y})$ and $S_{xx} = \sum_i (x_i - \bar{x})^2$. Moreover $\hat{a}$ and $\hat{b}$ are independent, with

$$\hat{a} \sim N\left(a, \frac{\sigma^2}{n}\right), \quad \hat{b} \sim N\left(b, \frac{\sigma^2}{S_{xx}}\right).$$

Proof. Minimise $S(a, b) = \sum_i (Y_i - a - b(x_i - \bar{x}))^2$. Since $\sum_i (x_i - \bar{x}) = 0$,

$$\frac{\partial S}{\partial a} = -2 \sum_i (Y_i - a - b(x_i - \bar{x})) = -2 \sum_i (Y_i - a),$$

which vanishes at $a = \bar{Y}$, independently of b . Substituting,

$$\frac{\partial S}{\partial b} = -2 \sum_i (Y_i - \bar{Y} - b(x_i - \bar{x})) (x_i - \bar{x}),$$

which vanishes at $b = S_{xY}/S_{xx}$.

For the distributions, the design matrix is $X = [\mathbf{1}, x - \bar{x}\mathbf{1}]$ whose columns are orthogonal, so

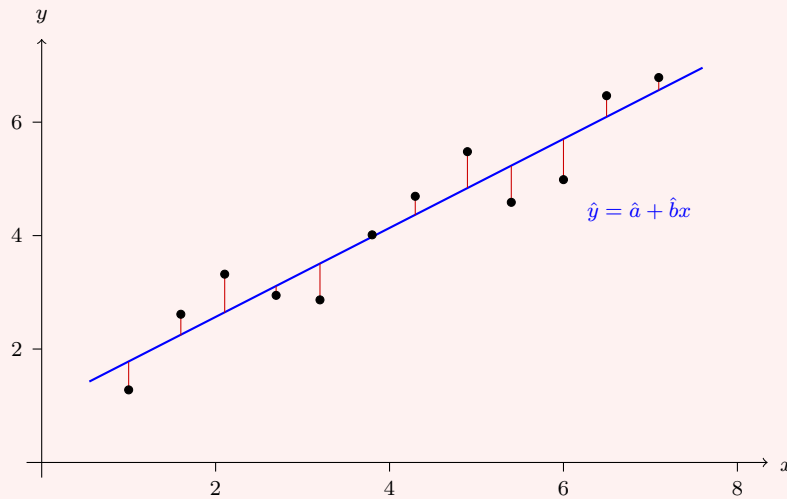
$$X^\top X = \begin{pmatrix} n & 0 \\ 0 & S_{xx} \end{pmatrix}, \quad (X^\top X)^{-1} = \begin{pmatrix} 1/n & 0 \\ 0 & 1/S_{xx} \end{pmatrix}.$$

By Theorem ??, $(\hat{a}, \hat{b})$ is bivariate normal with this covariance scaled by σ^2 ; the covariance is diagonal, so they are independent. $\square$

Note that S_{xY}/n is the sample covariance of x and Y , and S_{xx}/n the sample variance of x , so $\hat{b}$ is the sample covariance divided by the sample variance – exactly the formula for a regression coefficient one meets in elementary statistics.

Example 7.17 (A Worked Regression)

Twelve observations were taken, and the fitted line $\hat{y} = 0.99 + 0.79x$ obtained. The picture below shows the data, the fitted line, and the residuals as vertical segments.



Least squares chooses the line making the sum of the squared *vertical* red segments as small as possible. Note the asymmetry – vertical, not perpendicular, distance – which reflects the fact that the x values are treated as fixed and only Y is random.

§7.9 Prediction

Having fitted the model, we would like to use it. Let $x^* \in \mathbb{R}^p$ be a new vector of predictor values.

The **expected response** at x^* is $x^{*\top} \beta$, which we estimate by $x^{*\top} \hat{\beta}$. Since

$$x^{*\top} (\hat{\beta} - \beta) \sim N \left(0, \sigma^2 x^{*\top} (X^\top X)^{-1} x^* \right),$$

writing $\tau^2 = x^{*\top} (X^\top X)^{-1} x^*$ and dividing by the independent $\tilde{\sigma}$ as before,

$$\frac{x^{*\top} (\hat{\beta} - \beta)}{\tilde{\sigma} \tau} \sim t_{n-p}.$$

So a $100(1 - \alpha)\%$ confidence interval for the expected response is

$$x^{*\top} \hat{\beta} \pm \tilde{\sigma} \tau t_{n-p} \left(\frac{\alpha}{2} \right).$$

That, however, is not usually what we want. A new observation at x^* will be

$$Y^* = x^{*\top} \beta + \varepsilon^*, \quad \varepsilon^* \sim N(0, \sigma^2)$$

independent of the data, and there are now *two* sources of uncertainty: we do not know the line, and the new point will scatter about it anyway.

Definition 7.18 (Prediction Interval)

A $100(1 - \alpha)\%$ **prediction interval** for Y^* is a random interval $I(Y)$ with

$$\mathbb{P}(Y^* \in I(Y)) = 1 - \alpha,$$

where the probability is taken over the joint distribution of Y^* and $Y_1, \dots, Y_n$.

Proposition 7.19

$$x^{\star\top} \hat{\beta} \pm \tilde{\sigma} \sqrt{\tau^2 + 1} t_{n-p} \left(\frac{\alpha}{2} \right)$$

is a $100(1 - \alpha)\%$ prediction interval for Y^* .

Proof. Let $\hat{Y}^* = x^{\star\top} \hat{\beta}$. Then $\hat{Y}^* - Y^* = x^{\star\top} (\hat{\beta} - \beta) - \varepsilon^*$ has mean zero, and as the two terms are independent,

$$\text{Var}(\hat{Y}^* - Y^*) = \sigma^2 x^{\star\top} (X^\top X)^{-1} x^* + \sigma^2 = \sigma^2 (\tau^2 + 1).$$

Both terms are normal, so $\hat{Y}^* - Y^* \sim N(0, \sigma^2(\tau^2 + 1))$, and this is independent of $\tilde{\sigma}^2$. Dividing,

$$\frac{\hat{Y}^* - Y^*}{\tilde{\sigma} \sqrt{\tau^2 + 1}} \sim t_{n-p},$$

and rearranging gives the interval. □

The extra $+1$ under the square root is the irreducible noise of the new observation. Because τ^2 is typically small (of order $1/n$) while the 1 is not, prediction intervals are much wider than confidence intervals for the expected response, and they do *not* shrink to zero as $n \rightarrow \infty$. Confusing the two is a common and serious error.

Example 7.20

In the regression example above we have $n = 12$, $\bar{x} = 4.05$, $S_{xx} = 43.4$ and $\tilde{\sigma} = 0.54$, and we wish to predict at $x^* = 6$. With the centred parametrisation $x^{\star\top} = (1, x^* - \bar{x})$ and $(X^\top X)^{-1} = \text{diag}(1/n, 1/S_{xx})$, so

$$\tau^2 = \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} = \frac{1}{12} + \frac{(1.95)^2}{43.4} = 0.083 + 0.088 = 0.171,$$

giving $\tau = 0.41$. The fitted value is $0.99 + 0.79 \times 6 = 5.73$, and with $t_{10}(0.025) = 2.228$ we obtain a 95% confidence interval for the expected response

$$5.73 \pm 0.54 \times 0.41 \times 2.228 = (5.24, 6.22),$$

but a 95% prediction interval

$$5.73 \pm 0.54 \times \sqrt{1.171} \times 2.228 = (4.43, 7.03),$$

which is more than twice as wide. Notice also that τ^2 grows quadratically as x^* moves away from $\bar{x}$: extrapolation is punished, as it should be.

§7.10 One-Way Analysis of Variance

Our final application shows that the linear model covers situations that do not look like regression at all. Suppose we compare J treatments, with N subjects receiving each, and model

$$Y_{ij} = \alpha + \mu_j + \varepsilon_{ij}, \quad i = 1, \dots, N, \quad j = 1, \dots, J,$$

with ε_{ij} i.i.d $N(0, \sigma^2)$. Here the predictor is **categorical**: it records only which group a subject is in.

As written the parameters are not identifiable, since adding a constant to α and subtracting it from every μ_j changes nothing. We impose the **corner point constraint** $\mu_1 = 0$, so that group 1 (typically the control) is the baseline and μ_j measures the effect of treatment j relative to it. The design matrix is then

$$X = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & & \vdots \end{pmatrix} = (X_1 \ X_0)$$

where X_1 is the column of ones and X_0 the $J - 1$ indicator columns. Here $n = JN$ and $p = J$.

Definition 7.21 (One-Way ANOVA)

The **one-way analysis of variance** test is the test of

$$H_0 : \mu_2 = \cdots = \mu_J = 0, \quad H_1 : \mu_2, \dots, \mu_J \in \mathbb{R}$$

in the above model; under H_0 all groups have the same mean $\mathbb{E}[Y_{ij}] = \alpha$.

This is exactly the F -test of Section 7.7 with $p_0 = J - 1$. To write it out concretely we need only identify the two projections. Since $\text{col}(X)$ is the space of vectors constant within each group, P replaces every observation by its group mean:

$$\bar{Y}_{\cdot j} = \frac{1}{N} \sum_{i=1}^N Y_{ij}, \quad PY = (\bar{Y}_{\cdot 1}, \dots, \bar{Y}_{\cdot 1}, \dots, \bar{Y}_{\cdot J}, \dots, \bar{Y}_{\cdot J})^\top,$$

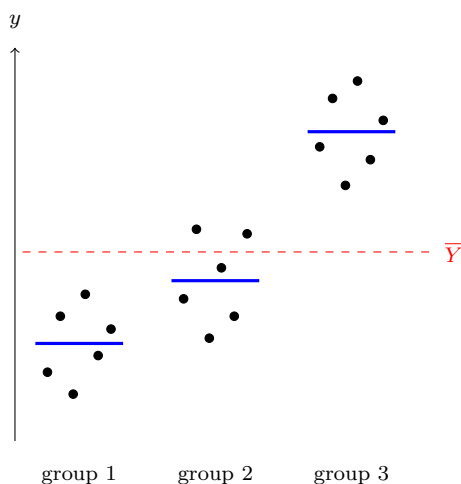
while $\text{col}(X_1)$ is the space of constant vectors, so P_1 replaces everything by the grand mean $\bar{Y}$. Therefore

$$\|(P - P_1)Y\|^2 = \sum_{j=1}^J N (\bar{Y}_{\cdot j} - \bar{Y})^2, \quad \|(I - P)Y\|^2 = \sum_{i,j} (Y_{ij} - \bar{Y}_{\cdot j})^2,$$

and the F -statistic is

$$F = \frac{\sum_j N (\bar{Y}_{\cdot j} - \bar{Y})^2 / (J - 1)}{\sum_{i,j} (Y_{ij} - \bar{Y}_{\cdot j})^2 / (JN - J)} = \frac{\text{variability between groups}}{\text{variability within groups}} \sim F_{J-1, JN-J}$$

under H_0 . This is where the name ‘analysis of variance’ comes from: the total sum of squares splits, by Pythagoras, into a between-groups part and a within-groups part, and we ask which dominates.



The blue bars are the group means $\bar{Y}_{.j}$ and the red dashed line is the grand mean. The F -statistic is large when the blue bars are spread out relative to the scatter of the dots about their own bar. In the picture above they arguably are, and indeed a formal test would reject.

Remark. Everything here generalises. **Two-way** analysis of variance handles groups classified by two factors,

$$Y_{ijk} = \alpha + \mu_j + \lambda_k + \varepsilon_{ijk},$$

again with corner point constraints $\mu_1 = \lambda_1 = 0$, and one may test $H_0 : \mu_j \equiv 0$, or $H_0 : \lambda_k \equiv 0$, or both, by exactly the same F -test machinery. One can also add *interaction* terms γ_{jk} to allow the effect of one factor to depend on the level of the other. Unequal group sizes cause no difficulty in principle, only messier formulae. In every case the recipe is the same: write down the design matrix, project, and compare sums of squares.

This is a good point at which to stop. The linear model, and its relatives, occupy a remarkable position: general enough to describe an enormous range of experiments, yet special enough that every distributional question about it can be answered exactly, by projecting onto a subspace and counting dimensions.